INCH-POUND

MIL-M-28787/4C(NW)
23 March 1994
SUPERSEDING
MIL-M-28787/4B(EC)
14 January 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, EXCLUSIVE-OR, DIGITAL, KEY CODE GDK

* This specification is approved for use by the Crane Division, Naval Surface Warfare Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic exclusive-OR gate module.

* 1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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* 3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

* 3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.186 failures per million operating hours for class I or 0.588 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

* 3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

* 3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

* 3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

* 3.6.1 Key code. The module key code shall be GDK.

* 3.6.2 Size. The module size shall be 1A.

* 3.6.3 Weight. The module shall have a maximum weight of 35 grams.

* 3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

* 3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

* 3.6.6 Depth. The module shall have no additional depth requirements.

* 3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

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* TABLE I Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	VOUT (low loaded)	III	S	10	800	10	800	10	800	10	800	10	800	mV
B1-B20	Input breakdown current	IV	S	0	90	0	90	0	90	0	90	0	90	µA
C1-C20	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D52	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E36	Output voltage, low level													
	Initial	VI	X	0	400	0	400	0	400	0	400	0	400	mV
	End-of-life	VI	X	0	420	0	440	0	440	0	440	0	440	mV
F1-F20	Input current, high level	VII	X	0	60	0	60	0	60	0	60	0	60	µA
* G1-G20	Input current, low level	VII	X	0.0	2.6	0.0	2.6	0.0	2.6	0.0	2.6	0.0	2.6	mA
* H1-H18	Output short circuit current	VII	S	20	220	20	220	20	220	20	220	20	220	mA
* J1	Power supply current, low level	VIII	S	50	175	50	175	50	175	50	175	50	175	mA
J2	Power supply current, high level	VIII	S	40	175	40	175	40	175	40	175	40	175	mA
K	Isolation													
	Initial	IX	S	10	INF	10	INF	10	INF	10	INF	10	INF	MOhm
	End-of-life	IX	S	1	INF	1	INF	1	INF	1	INF	1	INF	MOhm
* L1-L20	Input capacitance	X	Q	2	20	2	20	2	20	2	20	2	20	pF

- See footnotes at end of table.

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* TABLE I Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		80° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	M1-M9	Transition time (tTLH), (nine 2-input XOR)	XI	S	5	38	5	38	5	38	5	38	5	38	nS
*	M10,M12, M14,M16	Transition time (tTLH), (four 4-input XOR)	XI	S	5	38	5	38	5	38	5	38	5	38	nS
*	M11,M13, M15,M17	Transition time (tTLH), (four 4-input XOR)	XI	Q	5	38	5	38	5	38	5	38	5	38	nS
*	M18,M22	Transition time (tTLH), (two 8-input XOR)	XI	S	5	38	5	38	5	38	5	38	5	38	nS
*	M19-M21, M23-M25	Transition time (tTLH), (two 8-input XOR)	XI	S	5	38	5	38	5	38	5	38	5	38	nS
*	M26,M30	Transition time (tTLH), (two 8-input $\overline{XOR}$)	XI	S	5	38	5	38	5	38	5	38	5	38	nS
*	M27-M29, M31-M33	Transition time (tTLH), (two 8-input $\overline{XOR}$)	XI	Q	5	38	5	38	5	38	5	38	5	38	nS
*	M34	Transition time (tTLH), (one 2-input $\overline{XOR}$)	XI	S	5	38	5	38	5	38	5	38	5	38	nS
	N1-N9	Transition time (tTHL), (nine 2-input XOR)												
		Initial	XI	S	5	25	5	30	5	30	5	37	5	34	nS
		End-of-life	XI	S	5	27	5	32	5	32	5	39	5	36	nS

- See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N10,N12, N14,N16	Transition time (tTHL), (four 4-input XOR)													
	Initial	XI	S	5	25	5	30	5	30	5	37	5	34	nS
	End-of-life	XI	S	5	27	5	32	5	32	5	39	5	36	nS
N11,N13, N15,N17	Transition time (tTHL), (four 4-input XOR)													
	Initial	XI	Q	5	25	5	30	5	30	5	37	5	34	nS
	End-of-life	XI	Q	5	27	5	32	5	32	5	39	5	36	nS
N18,N22	Transition time (tTHL), (two 8-input XOR)													
	Initial	XI	S	5	25	5	30	5	30	5	37	5	34	nS
	End-of-life	XI	S	5	27	5	32	5	32	5	39	5	36	nS
N19-N21, N23-N25	Transition time (tTHL), (two 8-input XOR)													
	Initial	XI	Q	5	25	5	30	5	30	5	37	5	34	nS
	End-of-life	XI	Q	5	27	5	32	5	32	5	39	5	36	nS
N26,N30	Transition time (tTHL), (two 8-input $\overline{\text{XOR}}$)													
	Initial	XI	S	5	25	5	30	5	30	5	37	5	34	nS
	End-of-life	XI	S	5	27	5	32	5	32	5	39	5	36	nS
N27-N29, N31-N33	Transition time (tTHL), (two 8-input $\overline{\text{XOR}}$)													
	Initial	XI	Q	5	25	5	30	5	30	5	37	5	34	nS
	End-of-life	XI	Q	5	27	5	32	5	32	5	39	5	36	nS

- See footnotes at end of table.

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* TABLE I. Module electrical requirements Continued

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N34	Transition time (tTHL), (one 2-input $\overline{XOR}$)													
	Initial	XI	S	5	25	5	30	5	30	5	37	5	34	nS
	End-of-life	XI	S	5	27	5	32	5	32	5	39	5	36	nS
P1-P9	Propagation delay time (tPHL), one input at low level (nine 2-input XOR)													
	Initial	XI	S	7	27	7	33	7	33	7	33	7	33	nS
	End-of-life	XI	S	7	30	7	35	7	35	7	35	7	35	nS
P10-P17	Propagation delay time (tPHL), one input at low level (two 8-input XOR)													
	Initial	XI	S	14	54	14	66	14	66	14	66	14	66	nS
	End-of-life	XI	S	14	60	14	70	14	70	14	70	14	70	nS
P18-P25	Propagation delay time (tPHL), one input at low level (two 8-input XOR)													
	Initial	XI	S	21	81	21	99	21	98	21	99	21	99	nS
	End-of-life	XI	S	21	90	21	105	21	105	21	105	21	105	nS
P26-P33	Propagation delay time (tPHL), one input at low level (two 8-input $\overline{XOR}$)													
	Initial	XI	S	24	81	24	100	24	100	24	100	24	100	nS
	End-of-life	XI	S	24	90	24	108	24	108	24	108	24	108	nS

- See footnotes at end of table.

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* TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P34	Propagation delay time (tPHL), one input at low level (one 2-input $\overline{XOR}$)													
	Initial	XI	S	10	37	10	46	10	46	10	46	10	46	nS
	End-of-life	XI	S	10	40	10	48	10	48	10	48	10	48	nS
P35-P43	Propagation delay time (tPHL), one input at high level (nine 2-input XOR)													
	Initial	XI	S	3	23	3	29	3	29	3	29	3	29	nS
	End-of-life	XI	S	3	28	3	31	3	31	3	31	3	31	nS
P44-P51	Propagation delay time (tPHL), one input at high level (four 4-input XOR)													
	Initial	XI	S	6	41	6	53	6	53	6	53	6	53	nS
	End-of-life	XI	S	6	47	6	58	6	58	6	58	6	58	nS
P52-P59	Propagation delay time (tPHL), one input at high level (two 8-input XOR)													
	Initial	XI	S	9	64	9	82	9	82	9	82	9	82	nS
	End-of-life	XI	S	9	73	9	89	9	89	9	89	9	89	nS
P60-P67	Propagation delay time (tPHL), one input at high level (two 8-input $\overline{XOR}$)													
	Initial	XI	S	12	74	12	92	12	92	12	92	12	92	nS
	End-of-life	XI	S	12	83	12	99	12	99	12	99	12	99	nS

- See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P68	Propagation delay time (tPHL), one input at high level (one 2-input $\overline{XOR}$)													
	Initial	X1	S	6	33	6	39	6	39	6	39	6	39	nS
	End-of-life	X1	S	6	36	6	41	6	41	6	41	6	41	nS
Q1-Q9	Propagation delay time (tPLH), one input at low level (nine 2-input XOR)													
	Initial	X1	S	7	22	7	28	7	28	7	28	7	28	nS
	End-of-life	X1	S	7	25	7	30	7	30	7	30	7	30	nS
Q10-Q17	Propagation delay time (tPLH), one input at low level (four 4-input XOR)													
	Initial	X1	S	14	44	14	56	14	56	14	56	14	56	nS
	End-of-life	X1	S	14	50	14	60	14	60	14	60	14	60	nS
Q18-Q25	Propagation delay time (tPLH), one input at low level (two 8-input XOR)													
	Initial	X1	S	21	68	21	84	21	84	21	84	21	84	nS
	End-of-life	X1	S	21	75	21	90	21	90	21	90	21	90	nS
Q26-Q33	Propagation delay time (tPLH), one input at low level (two 8-input $\overline{XOR}$)													
	Initial	X1	S	24	91	24	112	24	112	24	112	24	112	nS
	End-of-life	X1	S	24	100	24	118	24	118	24	118	24	118	nS

- See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q34	Propagation delay time (tPLH), one input at low level (one 2-input $\overline{XOR}$)													
	Initial	XI	S	10	37	10	46	10	46	10	46	10	46	nS
	End-of-life	XI	S	10	40	10	48	10	48	10	48	10	48	nS
Q35-Q43	Propagation delay time (tPLH), one input at high level (nine 2-input XOR)													
	Initial	XI	S	3	22	3	28	3	28	3	28	3	28	nS
	End-of-life	XI	S	3	25	3	30	3	30	3	30	3	30	nS
Q44-Q51	Propagation delay time (tPLH), one input at high level (four 4-input XOR)													
	Initial	XI	S	6	41	6	53	6	53	6	53	6	53	nS
	End-of-life	XI	S	6	47	6	58	6	58	6	58	6	58	nS
Q52-Q59	Propagation delay time (tPLH), one input at high level (two 8-input XOR)													
	Initial	XI	S	9	59	9	84	9	84	9	84	9	84	nS
	End-of-life	XI	S	9	68	9	87	9	87	9	87	9	87	nS
Q60-Q67	Propagation delay time (tPLH), one input at high level (two 8-input $\overline{XOR}$)													
	Initial	XI	S	12	74	12	92	12	92	12	92	12	92	nS
	End-of-life	XI	S	12	83	12	99	12	99	12	99	12	99	nS

-See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Units
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q68	Propagation delay time [tPLH], one input at high level (one 2-input $\overline{XOR}$)													
	Initial	XI	S	6	33	6	39	6	39	6	39	6	39	nS
	End-of-life	XI	S	6	36	6	41	6	41	6	41	6	41	nS
R	Power supply transient amplitude	XII	Q	0	400	0	400	0	400	0	400	0	400	mV
S1-S18	Output crosstalk, high level	XIII	Q	0	400	0	400	0	400	0	400	0	400	mV
T1-T18	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply transient rise time	XIV	S	2.5	7 5	2.5	7 5	2.5	7.5	2.5	7.5	2 5	7.5	µS

1/ "X" parameters shall be 100 percent tested during the module acceptance tests

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

5/ XOR = exclusive-OR gate.

$\overline{XOR}$ = exclusive-NOR gate.

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* 3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/4-1 for class I and M28787/4-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

* 3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

* 3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

* 3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

* 3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 VDC, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 VDC ± 1 percent.

* 3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

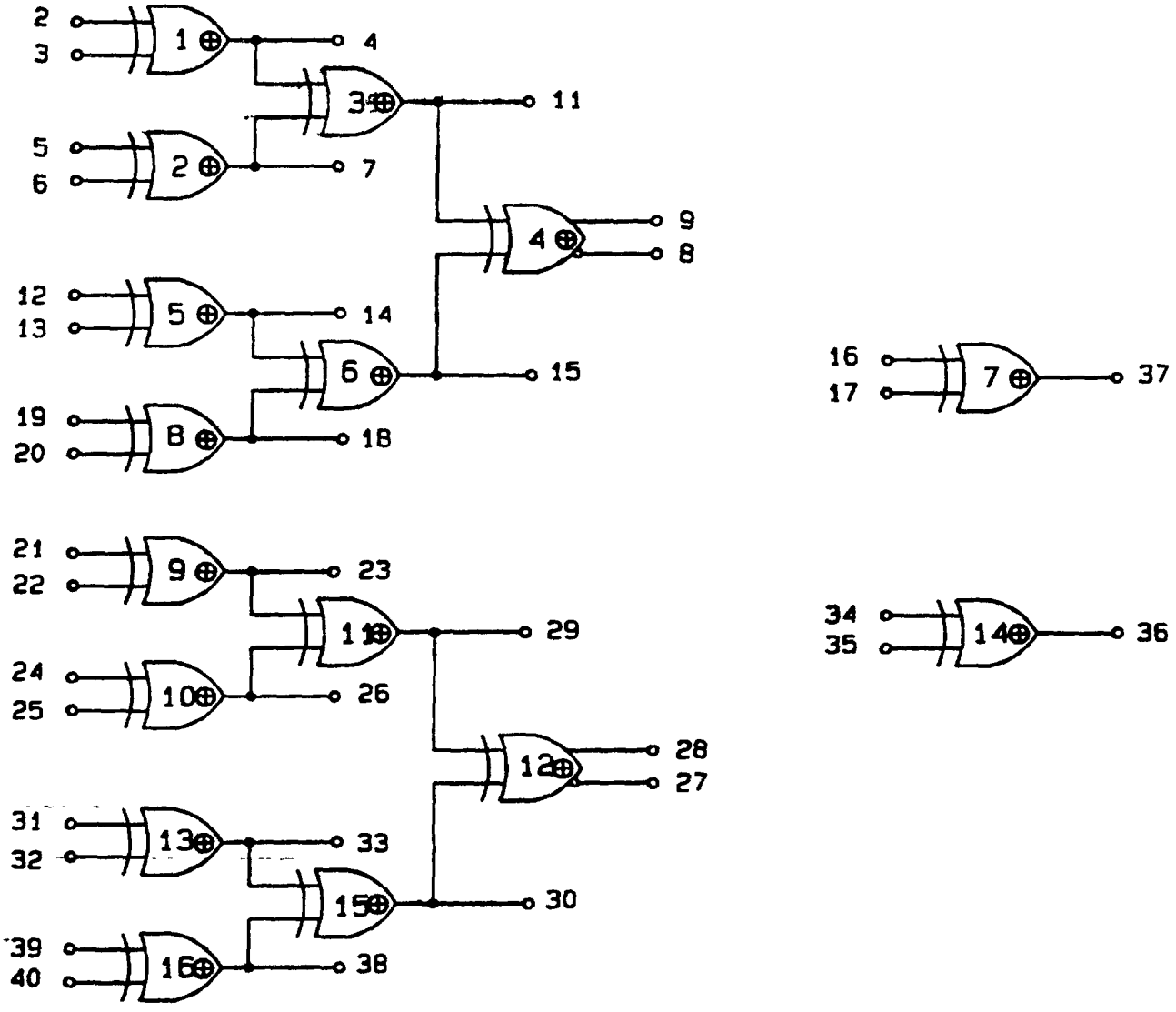
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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* FIGURE 1. GDK module function diagram, exclusive-OR gate.

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* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. Test conditions A, B, C, H and K are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, H or K are part of a sequence of tests, the order of testing shall be as follows: K, A, B, C and H where applicable, followed by the remaining tests in the sequence.

* 4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

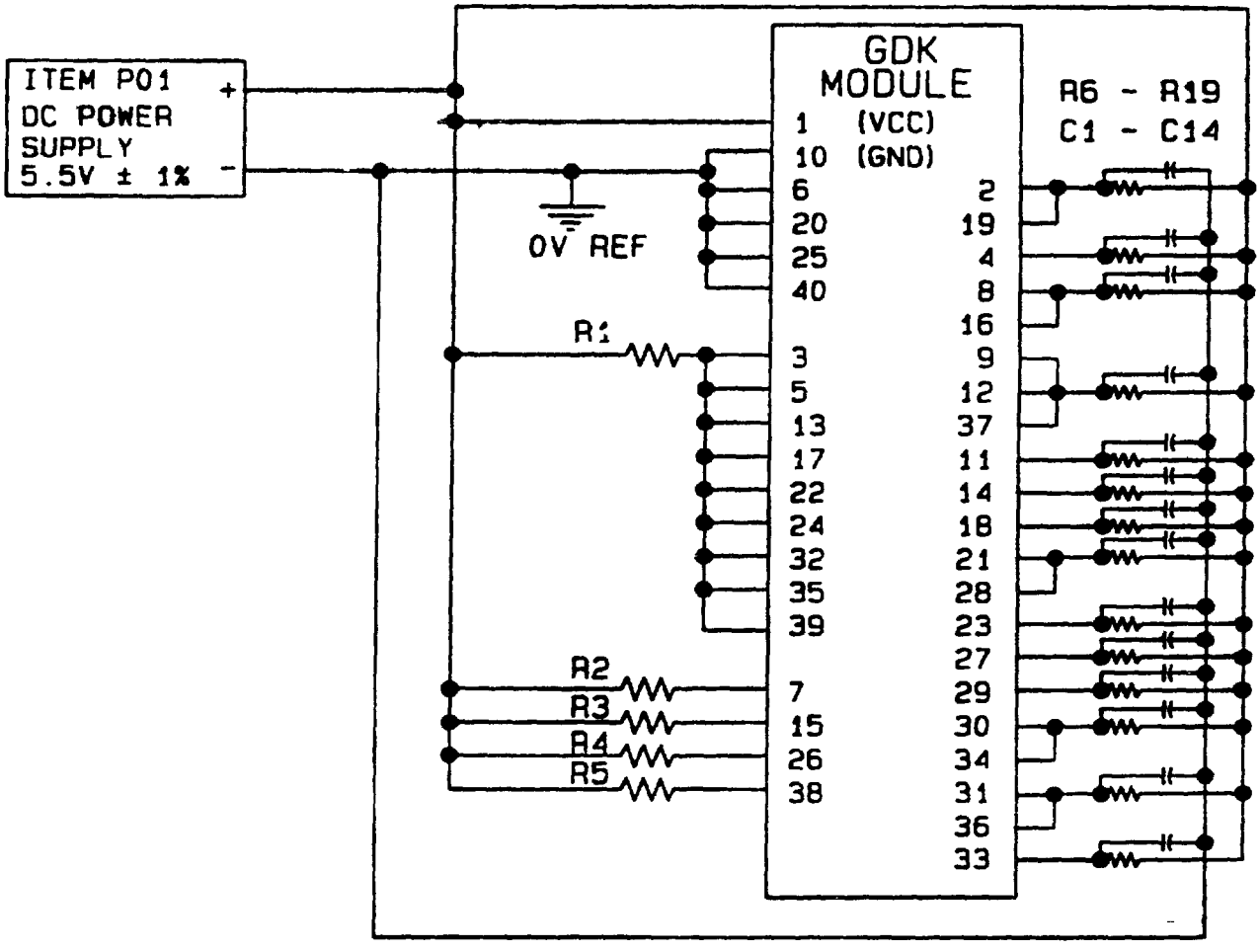
* 4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through G, J, M through Q, and U shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25 ± 5°C ambient.

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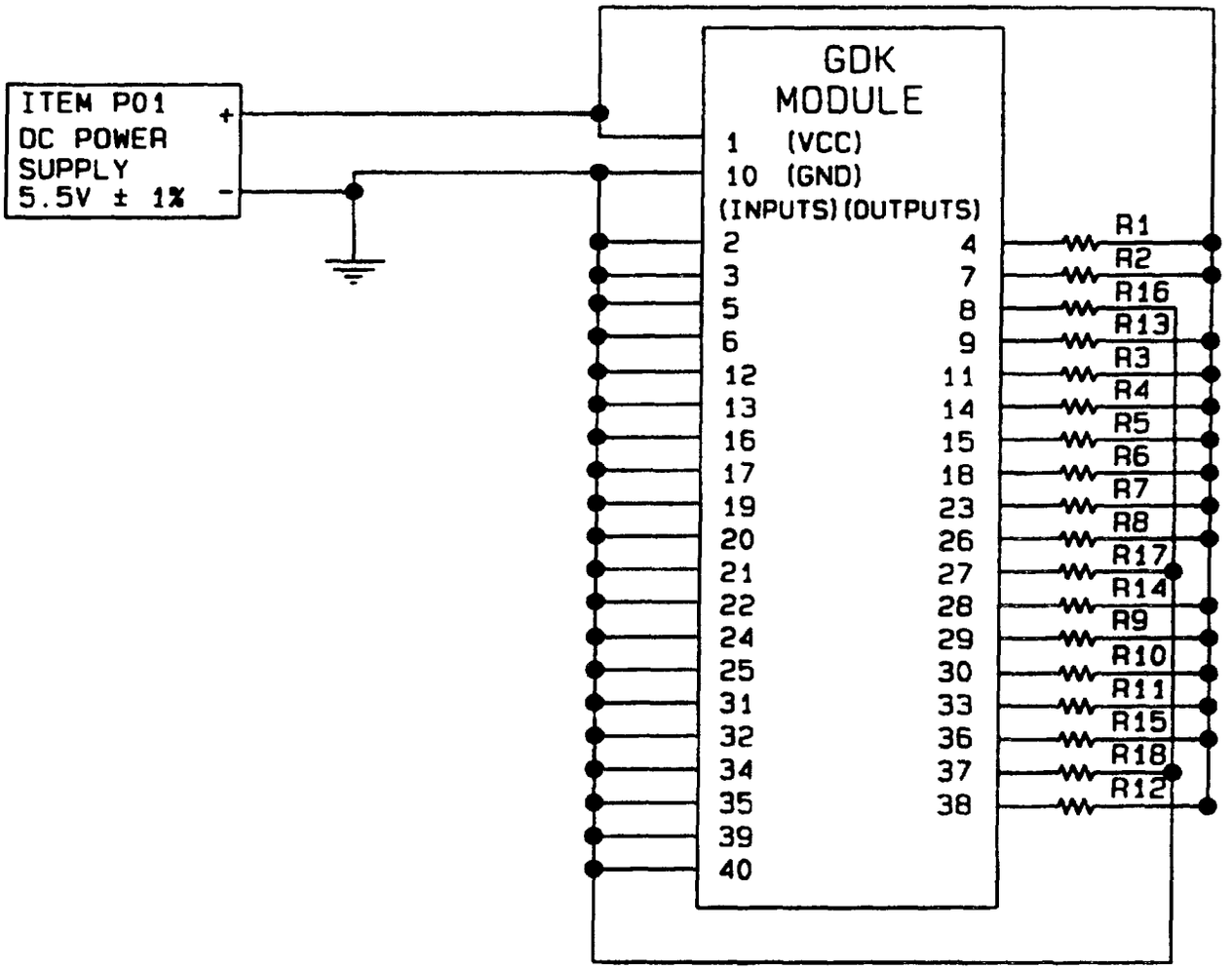


NOTES:

1. R1-R5 = 1.91 kOhms ± 1 per cent, 1/8 W.
2. R6-R19 = 374 Ohms ± 1 per cent, 1/8 W.
3. C1-C14 = 130 pF ± 20 per cent.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:

1. R1-R12 = 374 Ohms ± 1 per cent, 1/8 W.
2. R13-R15 = 316 Ohms ± 1 per cent, 1/8 W.
3. R16-R18 = 1.91 kOhms ± 1 per cent, 1/8 W.

FIGURE 3. GDK operating temperature test circuit.

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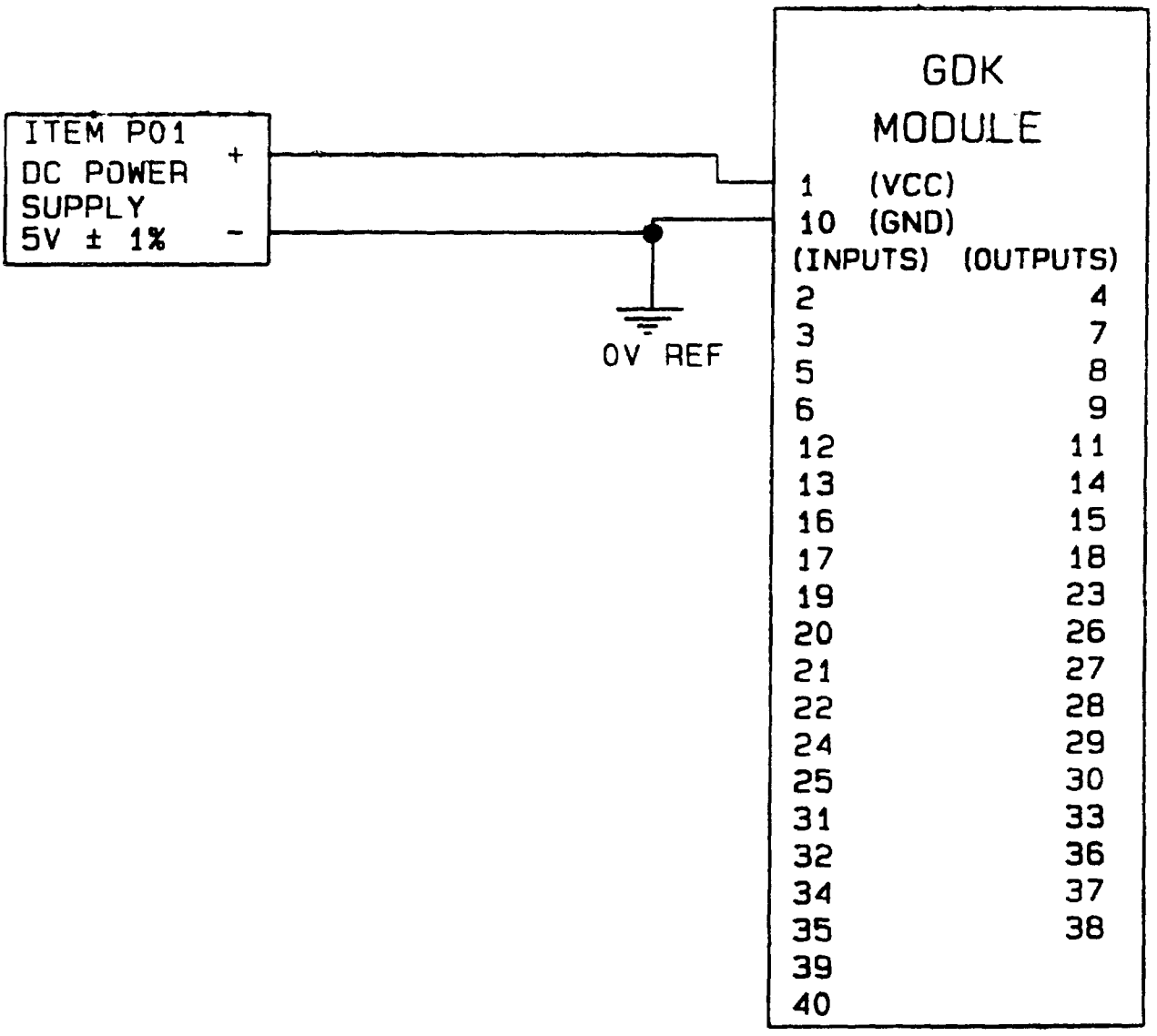


FIGURE 4. Vibration test circuit.

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* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through G, J and M through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests D through G, J, M through Q, and U shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

* 4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

* 6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/4 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658648. Part number M28787/4-1 supersedes part numbers 2658648 and M28787/4.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits in table I.

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	Class I	Class II	
Supply voltage	5 ± 10 per cent	5 ± 10 per cent	VDC
Maximum power dissipation			
All inputs at 0V	850 (outputs not loaded)	850 (outputs not loaded)	mW
One input at Vcc and other inputs at 0V	715 (outputs not loaded)	715 (outputs not loaded)	mW
Speed	DC to 10	DC to 10	MHz
Input voltage			
Low level	800 max	800 max	mV
High level	2 min	2 min	V
Input current			
Low level (Vil = 0V)	0.1 min, 2.6 max	0.1 min, 2.6 max	mA
High level (Vih = 2.4V)	0 min, 60 max	0 min, 60 max	μA
Output voltage			
Low level	400 max	400 max	mV
High level	2.4 min, Vcc max	2.4 min, 2.4 max	V
Output drive current			
Pin 8, 9, 27, 28, 3 6, or 37			
Low level	16 max	16 max	mA
High level	1.26 min	1.26 min	mA
Pin 4, 7, 11, 14, 1 5, 18, 23, 26, 29, 30, 33,or 38			
Low level	13.6 max	13.6 max	mA
High level	1.2 min	1.2 min	mA
Propagation delay time (tPHL)			
2-input XOR	7 min, 35 max	7 min, 35 max	nS
4-input XOR	14 min, 70 max	14 min, 70 max	nS
8-input XOR	21 min, 105 max	21 min, 105 max	nS
16-input XOR	140 max	140 max	nS
2-input $\overline{\text{XOR}}$	10 min, 48 max	10 min, 48 max	nS
8-input $\overline{\text{XOR}}$	24 min, 108 max	24 min, 108 max	nS
16-input $\overline{\text{XOR}}$	156 max	156 max	nS

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	Class I	Class II	
Propagation delay time (tPLH)			
2-input XOR	7 min, 30 max	7 min, 30 max	nS
4-input XOR	14 min, 60 max	14 min, 60 max	nS
8-input XOR	21 min, 90 max	21 min, 90 max	nS
16-input XOR	120 max	120 max	nS
2-input $\overline{\text{XOR}}$	10 min, 48 max	10 min, 48 max	nS
8-input $\overline{\text{XOR}}$	24 min, 118 max	24 min, 118 max	nS
16-input $\overline{\text{XOR}}$	138 max	138 max	nS

* 6.5 Module functions. This module contains the elements required to form any one of the following logic functions:

Nine 2-input exclusive-OR and one 2-input exclusive-NOR gates.

Four 4-input exclusive-OR, one 2-input exclusive-OR and one 2-input exclusive-NOR gates.

Two 8-input exclusive-OR, one 2-input exclusive-OR and one 2-input exclusive-NOR gates.

Two 8-input exclusive-NOR, one 2-input exclusive-OR and one 2-input exclusive-NOR gates.

One 16-input exclusive-OR.

One 16-input exclusive-NOR.

* 6.5.1 Input to output relationship. Each gate has the input to output relationship in table II for positive logic.

Each gate has either the Q or $\overline{Q}$ outputs or both in table II.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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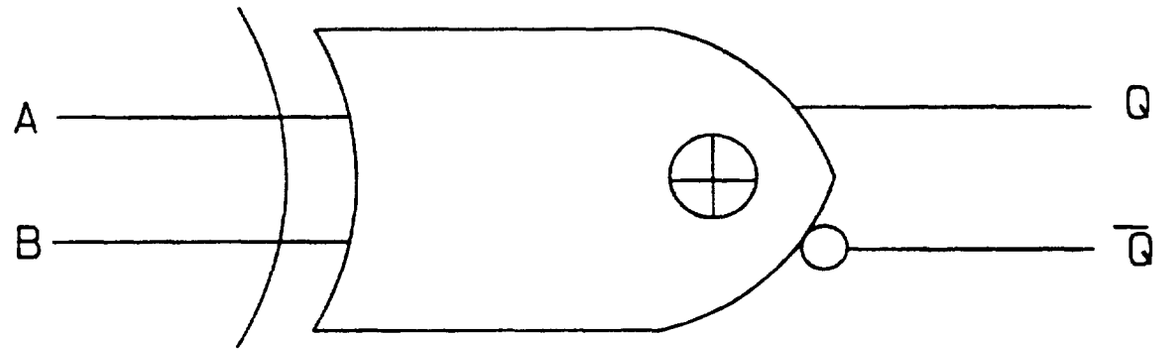
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TABLE II. Input to output relationship.



INPUTS	
A	B
L	L
L	H
H	L
H	H

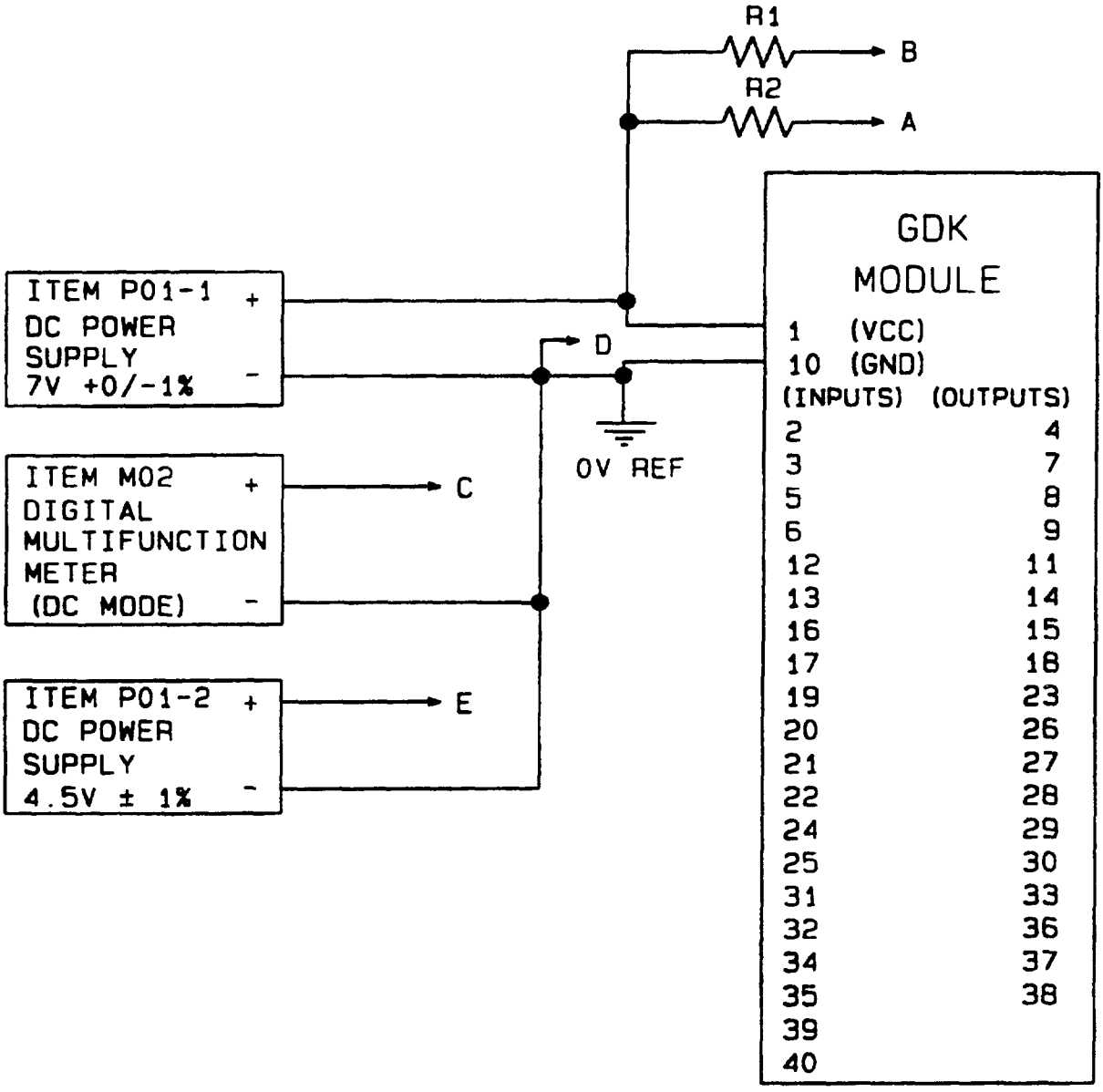
OUTPUTS	
Q	$\overline{Q}$
L	H
H	L
H	L
L	H

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* TABLE III. Vout (low-loaded) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
A1	Vout (low loaded)	4	Open	4	2	Open	10	800	mV	5
A2		7		7	Open					
A3		11	Open	11	2,11					
A4		Open	9	9	2	Open				
A5		Open	8	8	Open	2				
A6		14	Open	14	Open	Open				
A7		15	Open	15	Open	Open				
A8		Open	37	37	17	16				
A9		18	Open	18	Open	Open				
A10		23		23	21					
A11		26		26	Open					
A12		29		29	21					
A13		28	Open	28	21	Open				
A14		Open	27	27	Open	21				
A15		33	Open	33		Open				
A16		Open	36	36						
A17		30	Open	30						
A18	Vout (low loaded)	38	Open	38	Open	Open	10	800	mV	5

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NOTES:

1. R1 = 316 Ohms ± 1 per cent, 1/8 W.
2. R2 = 374 Ohms ± 1 per cent, 1/8 W.
3. After point A or point B has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.

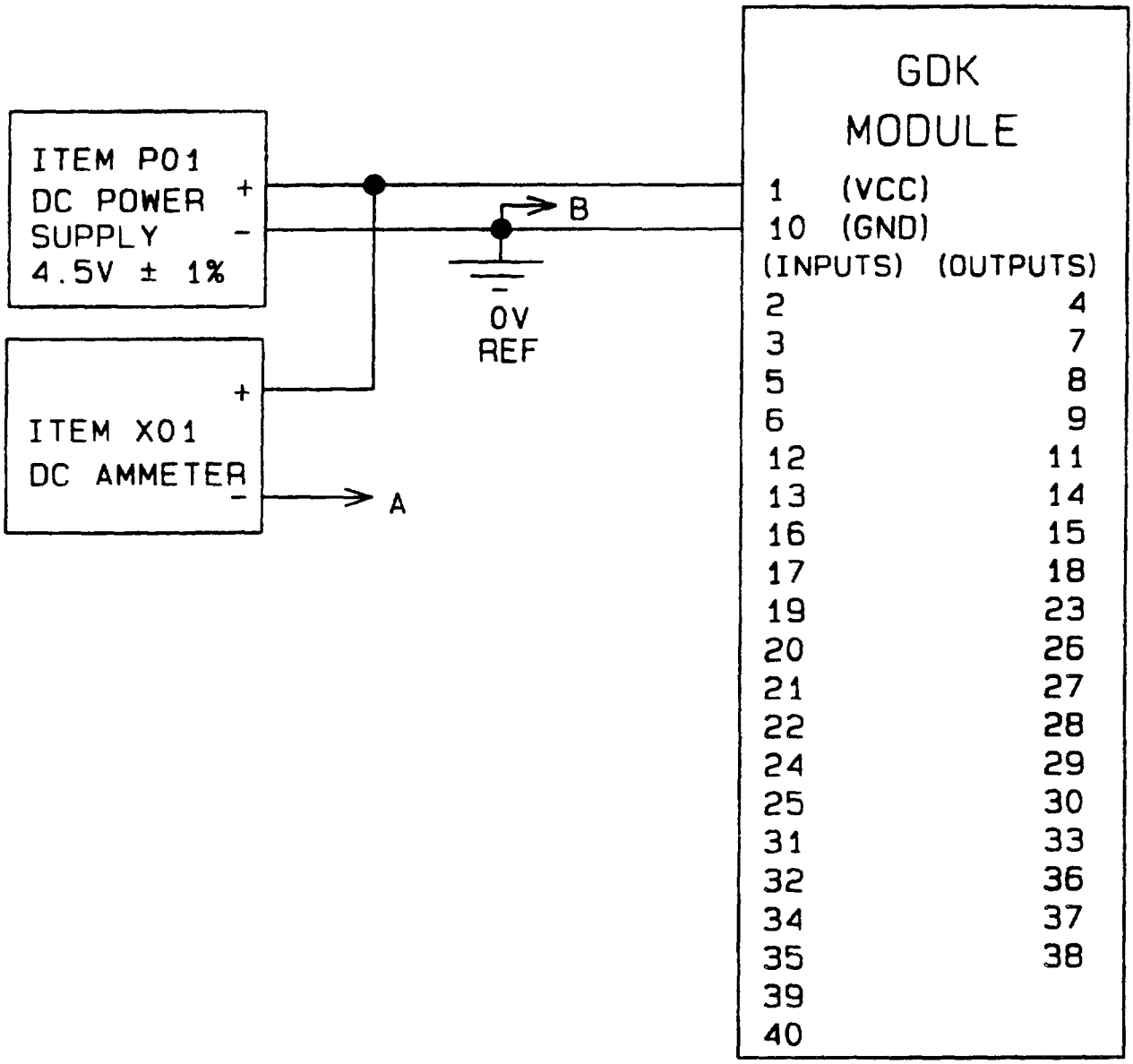
* FIGURE 5. Vout (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test cond	Parameter	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
B1	Input breakdown current	2	3	0	90	µA	6
B2		3	2				
B3		5	6				
B4		6	5				
B5		12	13				
B6		13	12				
B7		16	17				
B8		17	16				
B9		19	20				
B10		20	19				
B11		21	22				
B12		22	21				
B13		24	25				
B14		25	24				
B15		31	32				
B16		32	31				
B17		34	35				
B18		35	34				
B19		39	40				
B20	Input breakdown current	40	39	0	90	µA	6

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CAUTION: **Do not isolate the chassis of item X01 from earth ground.**

NOTE: **Input pins 2, 3, 5, 6, 12, 13, 16, 17, 19, 20, 21, 22, 24, 25, 31, 32, 34, 35, 39 and 40 shall be connected to point B, except when connected to point A as specified in table IV.**

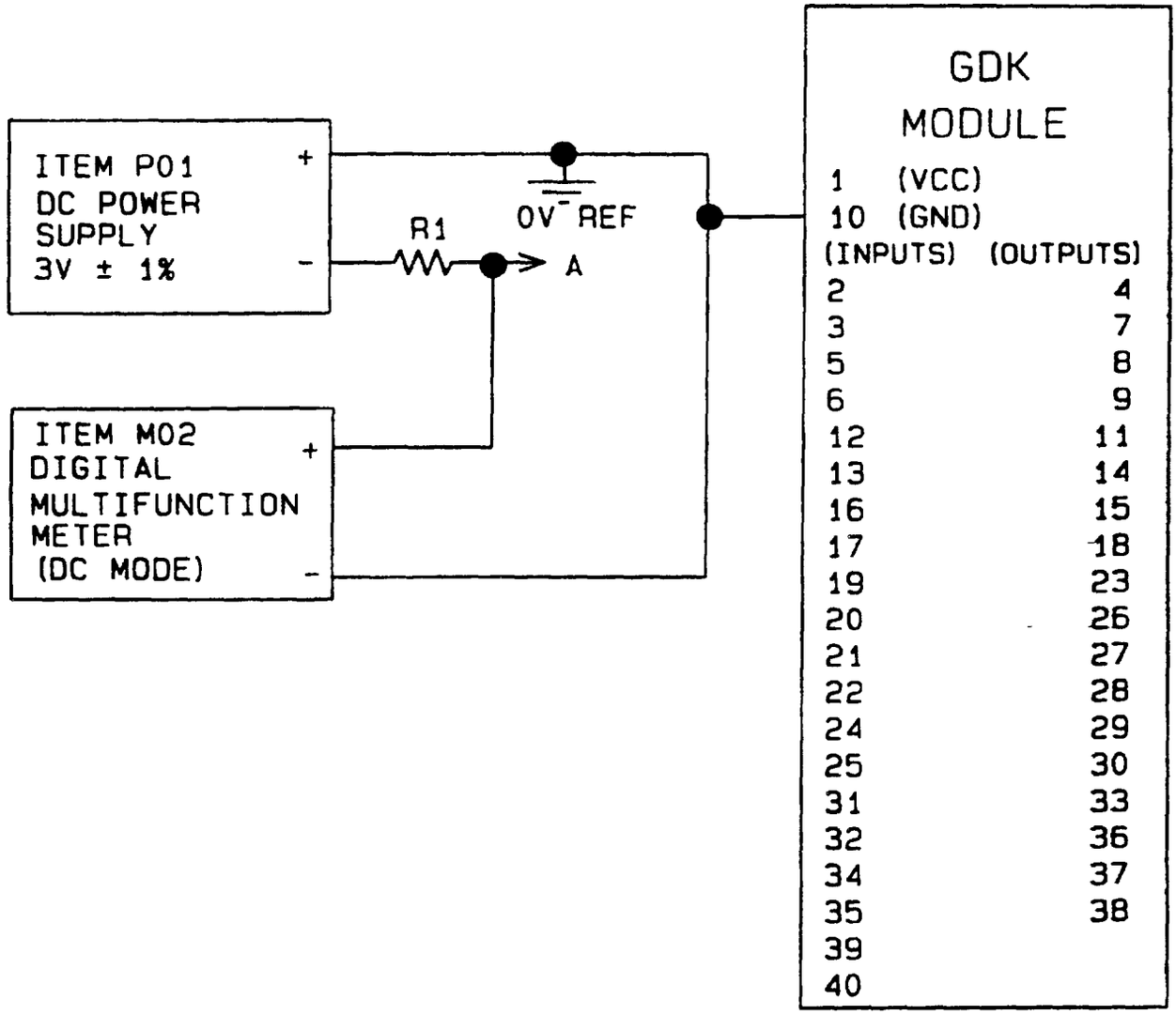
* **FIGURE 6.** **Input breakdown current test circuit for table IV.**

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TABLE V. Input clamp diode voltage test.

Test cond	Parameter	Connect A to pin	25 ± 5°C Initial limits Min	Max	Unit	Test fig
C1	Input clamp diode voltage	2	-0.4	-1.5	V	7
C2		3				
C3		5				
C4		6				
C5		12				
C6		13				
C7		16				
C8		17				
C9		19				
C10		20				
C11		21				
C12		22				
C13		24				
C14		25				
C15		31				
C16		32				
C17		34				
C18		35				
C19		39				
C20	Input clamp diode voltage	40	-0.4	-1.5	V	7

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NOTE: R1 = 150 Ohms ± 1 per cent, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

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* TABLE VI. Output voltage (high level and low level) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
D1	Output voltage, high level (input threshold, high level)	2	4	Open	3	2.4	4.5	V	8
D2		3	4		2				
D3		5	7		6				
D4		6	7		5				
D5		12	14		13				
D6		13	14		12				
D7		19	18		20				
D8		20	18		19				
D9		21	23		22				
D10		22	23		21				
D11		24	26		25				
D12		25	26		24				
D13		31	33		32				
D14		32	33		31				
D15		34	36		35				
D16		35	36		34				
D17		39	38		40				
D18	Output voltage, high level (input threshold, high level)	40	38	Open	39	2.4	4.5	V	8

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* TABLE VI. Output voltage (high level and low level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
D19	Output voltage, high level (input threshold, high level)	16, 17	37	Open	Open	2.4	4.5	V	8
D20	Output voltage, high level (input threshold, low level)	2	4	3					
D21		3	4	2					
D22		5	7	6					
D23		6	7	5					
D24		12	14	13					
D25		13	14	12					
D26		19	18	20					
D27		20	18	19					
D28		21	23	22					
D29		22	23	21					
D30		24	26	25					
D31		25	26	24					
D32		31	33	32					
D33		32	33	31					
D34	Output voltage, high level (input threshold, low level)	34	36	35	Open	2.4	4.5	V	8

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* TABLE VI. Output voltage (high level and low level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
D35	Output voltage, high level (input threshold, low level)	35	36	34	Open	2.4	4.5	V	8
D36		39	38	40					
D37		40	38	39					
D38	Output voltage, high level (input threshold, low level)	16, 17	37	Open	Open				
D39	Output voltage, high level	Open	11	2	3,5, 6				
D40			11	2	3,5,6				
D41			15	12	13,19, 20				
D42			15	19	12,13, 20				
D43			29	21	22,24, 25				
D44			29	24	21,22, 25				
D45			30	31	32,39, 40				
D46	Output voltage, high level	Open	30	39	31,32, 40	2.4	4.5	V	8

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* TABLE VI. Output voltage (high level and low level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
D47	Output voltage, high level	Open	9	2	3,5,6, 12,13, 19,20	2.4	4.5	V	8
D48			9	12	2,3,5, 6,13, 19,20				
D49			28	21	22,24, 25,31, 32,39, 40				
D50			28	31	21,22, 24,25, 32,39, 40				
D51			8	Open	2,3,5, 6,12, 13,19, 20				
D52	Output voltage, high level	Open	27		21,22, 24,25, 31,32, 39,40	2.4	4.5	V	
E1	Output voltage, low level (input threshold, high level)	16	37		17	0	400	mV	
E2	Output voltage, low level (input threshold, high level)	17	37	Open	16	0	400	mV	8

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* TABLE VI. Output voltage (high level and low level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
E3	Output voltage, low level (input threshold, high level)	2,3	4	Open	Open	0	400	mV	8
E4		5,6	7						
E5		12,13	14						
E6		19,20	18						
E7		21,22	23						
E8		24,25	26						
E9		31,32	33						
E10		34,35	36						
E11	Output voltage, low level (input threshold, high level)	39,40	38	Open					
E12	Output voltage, low level (input threshold, low level)	16	37	17					
E13	Output voltage, low level (input threshold, low level)	17	37	16	Open	0	400	mV	8

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* TABLE VI. Output voltage (high level and low level) test - Continued.

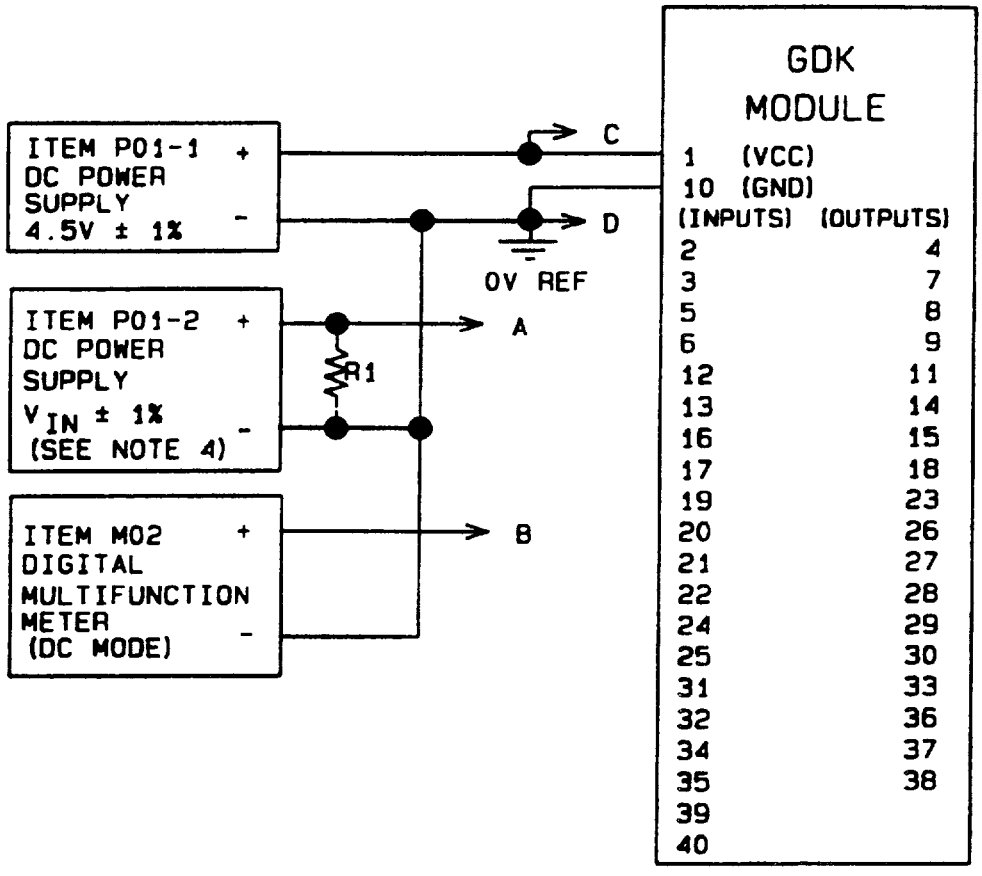
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
E14	Output voltage, low level (input threshold, low level)	2,3	4	Open	Open	0	400	mV	8
E15		5,6	7						
E16		12,13	14						
E17		19,20	18						
E18		21,22	23						
E19		24,25	26						
* E20		31,32	33						
E21		34,35	36						
E22	Output voltage, low level (input threshold, low level)	39,40	38	Open	Open				
E23	Output voltage, low level	Open	11	2,5	3,6				
E24			15	12, 19	13,20				
E25	Output voltage, low level	Open	29	21, 24	22,25	0	400	mV	8

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* TABLE VI. Output voltage (high level and low level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
E26	Output voltage, low level	Open	30	31, 39	32,40	0	400	mV	8
E27			11	Open	2,3,5, 6				
E28			15		12,13, 19,20				
E29			29		21,22, 24,25				
E30			30	Open	31,32, 39,40				
E31			8	2	3,5,6, 12,13, 19,20				
E32			27	21	22,24, 25,31, 32,39, 40				
E33			9	2,12	3,5,6, 13,19, 20				
E34			28	21, 31	22,24, 25,32, 39,40				
E35			9	Open	2,3,5, 6,12, 13,19, 20				
E36	Output voltage, low level	Open	28	Open	21,22, 24,25, 31,32, 39,40	0	400	mV	8

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NOTES:

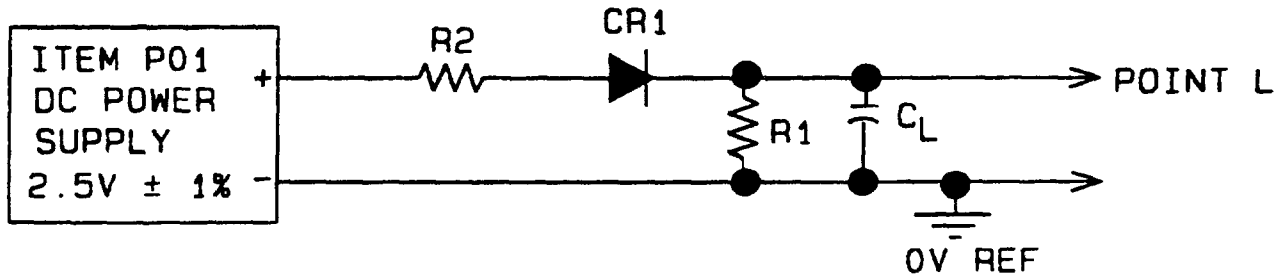
1. R1 = 50 Ohms ± 5 per cent, 1 W.
2. Connect load circuit number 1 (figure 9) to pins 8, 9, 27, 28, 36 and 37.
3. Connect load circuit number 2 (figure 9) to pins 4, 7, 11, 14, 15, 18, 23, 26, 29, 30, 33 and 38.
*4. Set item P01-2 voltage "Vin" as follows:

Test condition		Unit
D1-D19, E1-E11	2.0	V
D20-D38, E12-E22	0.8	V
D39-D52, E23-E36	No connection	

*5. Input pins 2, 3, 5, 6, 12, 13, 16, 17, 19, 20, 21, 22, 24, 25, 31, 32, 34, 35, 39 and 40 shall be connected to point C, except when specified otherwise on table VI.

* FIGURE 8. Output voltage (high level and low level) test circuit for table VI.

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NOTES:

1. Load circuit number 1 parameters:
   (a) CR1 = 1N4150 or equivalent.
   (b) CL = 140 ± 5 pF (including probe and parasitic capacitance).
   (c) R1 = 1.91 kOhms ± 1 per cent, 1/8 W.
   (d) R2 = Select value for a current flow of 16 mA ± 1 per cent out of point L with point L held at 0.4V.

2. Load circuit number 2 parameters:
   (a) CR1 = 1N4150 or equivalent.
   (b) CL = 130 ± 5 pF (including probe and parasitic capacitance).
   (c) R1 = 2 kOhms ± 1 per cent, 1/8 W.
   (d) R2 = Select value for a current flow of 13.6 mA ± 1 per cent out of point L with point L held at 0.4V.

3. Capacitor CL is not required for test conditions D and E.

FIGURE 9. GDK logic load circuit.

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* TABLE VII. Input current (high level and low level) and output short circuit current test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
F1	Input current, high level	2	Open	3	0	60	µA	10
F2		3		2				
F3		5		6				
F4		6		5				
F5		12		13				
F6		13		12				
F7		16		17				
F8		17		16				
F9		19		20				
F10		20		19				
F11		21		22				
F12		22		21				
F13		24		25				
F14		25		24				
F15		31		32				
F16		32		31				
F17		34		35				
F18		35		34				
F19		39		40				
F20	Input current, high level	40	Open	39	0	60	µA	10

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* TABLE VII. Input current (high level and low level) and output short circuit current test - Continued.

	Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
*	G1	Input current, low level	3	2	Open	0.0	2.6	mA	10
	G2		2	3					
	G3		6	5					
	G4		5	6					
	G5		13	12					
	G6		12	13					
	G7		17	16					
	G8		16	17					
	G9		20	19					
	G10		19	20					
	G11		22	21					
	G12		21	22					
	G13		25	24					
	G14		24	25					
	G15		32	31					
	G16		31	32					
	G17		35	34					
	G18		34	35					
	G19		40	39					
	G20	Input current, low level	39	40	Open	0.0	2.6	mA	10

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* TABLE VII. Input current (high level and low level) and output short circuit current test - Continued

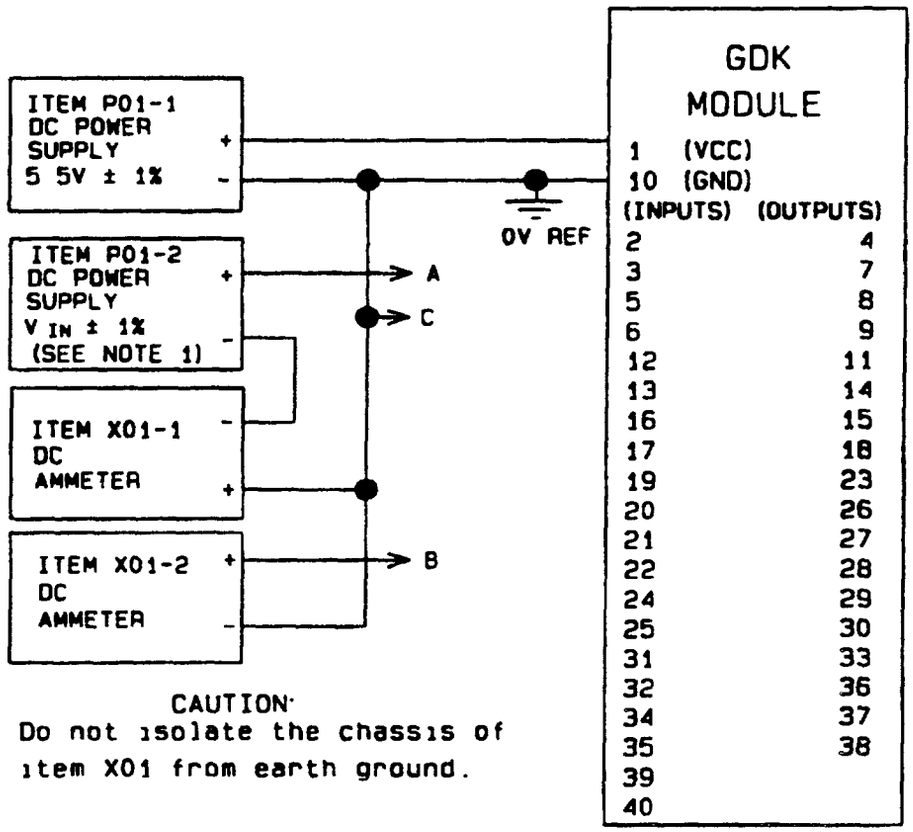
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
* H1	Output short circuit current	2	4	3	20	220	mA	10
H2		5	7	6				
H3		2	11	3,5,6				
H4		2	9	3,5,6, 12,13, 19,20				
H5		Open	8	2,3,5,6, 12,13, 19,20				
H6		12	14	13				
H7		12	15	13,19,20				
H8		Open	37	16,17				
H9		19	18	20				
H10		21	23	22				
H11		24	26	25				
H12		21	29	22,24,25				
H13		21	28	22,24, 25,31, 32,39,40				
H14		Open	27	21,22, 24,25, 31,32, 39,40				
H15	Output short circuit current	31	33	32	20	220	mA	10

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* TABLE VII. Input current (high level and low level) and output short circuit current test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
H16	Output short circuit current	34	36	35	20	220	mA	10
H17	"	31	30	32,39,40	"	"	"	"
H18	Output short circuit current	39	38	40	20	220	mA	10

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTES:

1. For test condition F, adjust Vin = 2.4V. For test conditions G and H, adjust Vin = 4.5V.

2. Use item X01-1 for test condition F and item X01-2 for test conditions G and H.

3. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

*4. For test condition H, input pins 2, 3, 5, 6, 12, 13, 16, 17, 19, 20, 21, 22, 24, 25, 31, 32, 34, 35, 39 and 40 shall be connected to point A, except when otherwise specified in table VII.

5. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within 5 seconds.

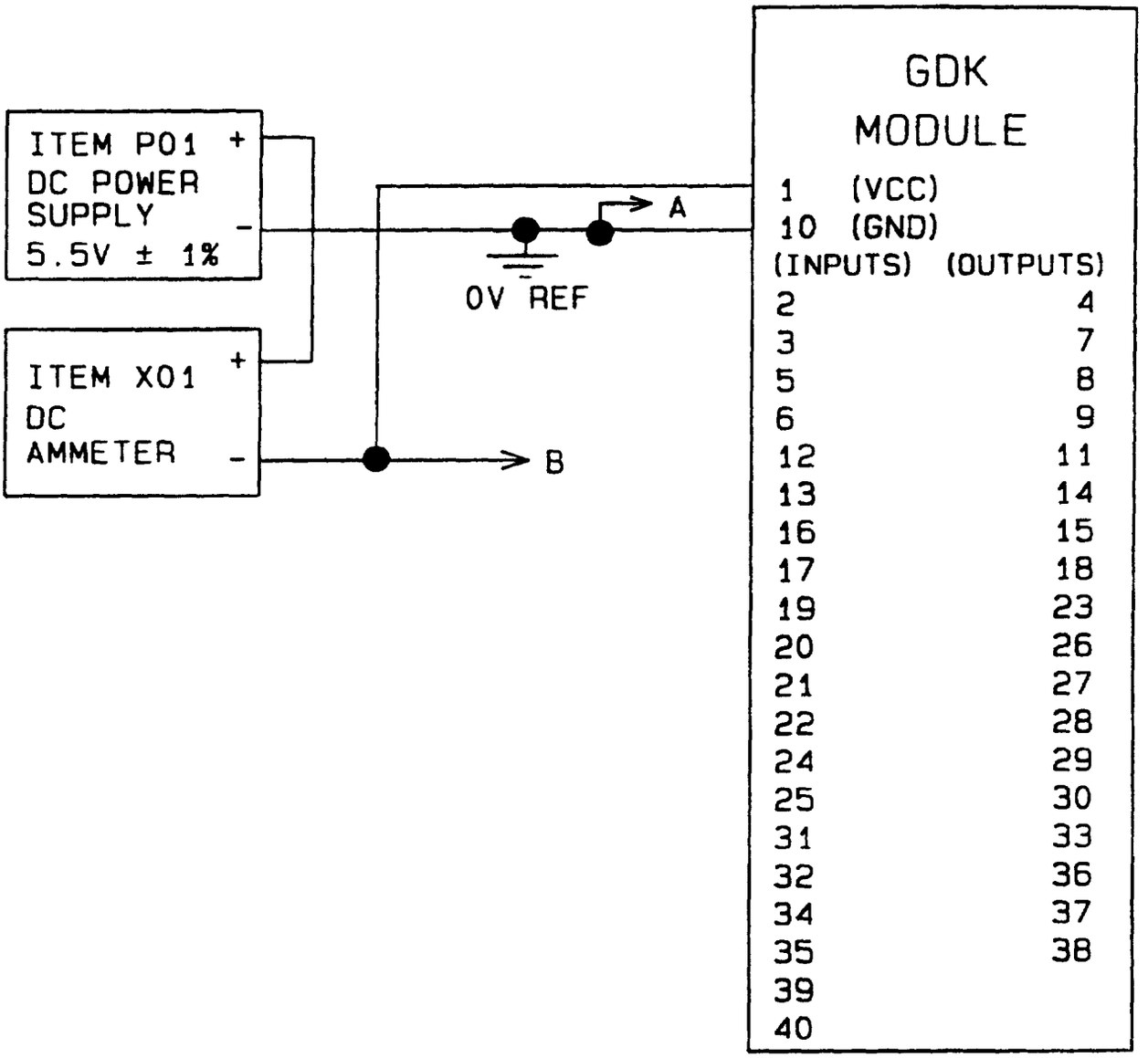
* FIGURE 10. Input current (high level and low level) and output short circuit current test circuit for table VII.

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* TABLE VIII. Power supply current (low level and high level) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
J1	Power supply current, low level (all inputs at ground)	2,3,5,6, 12,13, 16,17, 19,20, 21,22, 24,25, 31,32, 34,35, 39,40	Open	50	175	mA	11
J2	Power supply current, high level (one input at Vcc, other input at ground)	3,6,13, 17,20, 22,25, 32,35,40	2,5,12, 16,19, 21,24, 31,34, 39	40	175	mA	11

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CAUTION: **Do not isolate the chassis of item X01 from earth ground.**

FIGURE 11. **Power supply current (low level and high level) test circuit for table VIII.**

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TABLE IX. Isolation test.

Test cond	Parameter	Connect megohm bridge ITEM R06		Initial limits 25° ± 5°C ambient		Unit	Test fig
		+ Terminal	- Terminal	Min	Max		
K	Isolation	All pins	1/ Module frame	10	INF	MOhm	-

1/ Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal. Item R06 output shall be set at 100 ± VDC.

WARNING

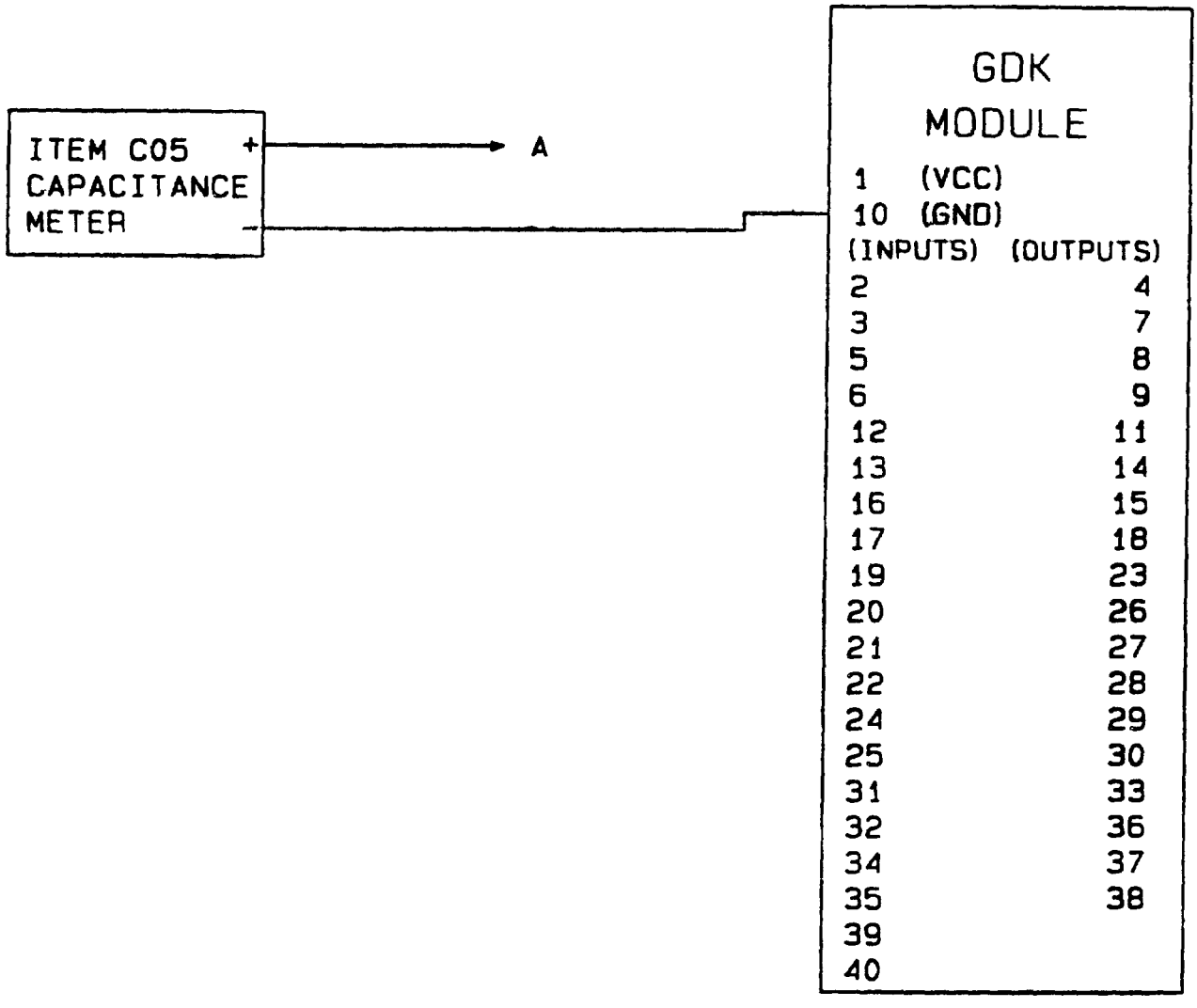
When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "DISCHARGE" before connecting or disconnecting the test leads.

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* TABLE X. Input capacitance test.

	Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C ambient		Unit	Test fig
				Min	Max		
*	L1	Input capacitance	2	2	20	pF	12
	L2		3				
	L3		5				
	L4		6				
	L5		12				
	L6		13				
	L7		16				
	L8		17				
	L9		19				
	L10		20				
	L11		21				
	L12		22				
	L13		24				
	L14		25				
	L15		31				
	L16		32				
	L17		34				
	L18		35				
	L19		39				
	L20	Input capacitance	40	2	20	pF	12

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NOTES:

1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the reading.
2. Null out the effects of the test lead prior to connecting the module.

FIGURE 12. Input capacitance test circuit for table X.

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient tTLH Min	Initial limits 25° ± 5°C ambient tTLH Max	Initial limits 25° ± 5°C ambient tTHL Min	Initial limits 25° ± 5°C ambient tTHL Max	Unit	Test fig
* M1,N1	Transition time	2	4	Open	3	5	38	5	25	nS	13, 14
M2,N2		5	7		6						
M3,N3		12	14		13						
M4,N4		19	18		20						
M5,N5		21	23		22						
M6,N6		24	26		25						
M7,N7		31	33		32						
M8,N8		34	36		35						
M9,N9		39	38		40						
M10,N10		2	11		3,5,6						
M11,N11		5	11		2,3,6						
M12,N12		12	15		13,19,20						
M13,N13		19	15		12,13,20						
M14,N14	Transition time	21	29	Open	22,24,25	5	38	5	25	nS	13, 14

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C ambient		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	tTLH		tTHL			
						Min	Max	Min	Max		
M15,N15	Transition time	24	29	Open	21,22,25	5	38	5	25	nS	13, 14
M16,N16		31	30		32,39,40						
M17,N17		39	30		31,32,40						
M18,N18		2	9		3,5,6, 12,13, 19,20						
M19,N19		5			2,3,6, 12,13, 19,20						
M20,N20		12			2,3,5,6, 12,19,20						
M21,N21		19	9		2,3,5,6, 12,13,20						
M22,N22		21	28		22,24, 25,31, 32,39,40						
M23,N23	Transition time	24	28	Open	21,22, 25,31, 32,39,40	5	38	5	25	nS	13, 14

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C ambient		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	tTLH		tTHL			
						Min	Max	Min	Max		
M24,N24	Transition time	31	28	Open	21,22, 24,25, 32,39,40	5	38	5	25	nS	13, 14
M25,N25		39	28		21,22, 24,25, 31,32,40						13, 14
M26,N26		2	8		3,5,6, 12,13, 19,20						13, 15
M27,N27		5			2,3,6, 12,13, 19,20						
M28,N28		12			2,3,5,6, 13,19,20						
M29,N29		19	8		2,3,5,6, 12,13,20						
M30,N30	Transition time	21	27	Open	22,24, 25,31, 32,39,40	5	38	5	25	nS	13, 15

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C ambient		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	tTLH		tTHL			
						Min	Max	Min	Max		
M31,N31	Transition time	24	27	Open	21,22, 25,31, 32,39,40	5	38	5	25	nS	13, 15
M32,N32		31			21,22, 24,25, 32,39,40						
M33,N33		39	27		21,22, 24,25, 31,32,40						
M34,N34	Transition time	16	37		17	5	38	5	25		13, 15
P1,Q1	Propagation delay time	2	4		3	7	27	7	22		13, 14
P2,Q2		5	7		6						
P3,Q3		12	14		13						
P4,Q4		19	18		20						
P5,Q5		21	23		22						
P6,Q6		24	26		25						
P7,Q7	Propagation delay time	31	33	Open	32	7	27	7	22	nS	13, 14

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* TABLE XI Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient tTLH Min	Initial limits 25° ± 5°C ambient tTLH Max	Initial limits 25° ± 5°C ambient tTHL Min	Initial limits 25° ± 5°C ambient tTHL Max	Unit	Test fig
P8,Q8	Propagation delay time	34	36	Open	35	7	27	7	22	nS	13, 14
P9,Q9		39	38		40	7	27	7	22		
P10,Q10		2	11		3,5,6	14	54	14	44		
P11,Q11		5	11		2,3,6						
P12,Q12		12	15		13,19,20						
P13,Q13		19	15		12,13,20						
P14,Q14		21	29		22,24,25						
P15,Q15		24	29		21,22,25						
P16,Q16		31	30		32,39,40						
P17,Q17		39	30		31,32,40	14	54	14	44		
P18,Q18		2	9		3,5,6, 12,13, 19,20	21	81	21	66		
P19,Q19	Propagation delay time	5	9	Open	2,3,6, 12,13, 19,20	21	81	21	66	nS	13, 14

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient tTLH Min	Initial limits 25° ± 5°C ambient tTLH Max	Initial limits 25° ± 5°C ambient tTHL Min	Initial limits 25° ± 5°C ambient tTHL Max	Unit	Test fig
P20,Q20	Propagation delay time	12	9	Open	2,3,5,6, 13,19,20	21	81	21	66	nS	13, 14
P21,Q21		19	9		2,3,5,6, 12,13,20						
P22,Q22		21	28		22,24, 25,31, 32,39,40						
P23,Q23		24			21,22, 25,31, 32,39,40						
P24,Q24		31			21,22, 24,25, 32,39,40						
P25,Q25		39	28		21,22, 24,25, 31,32,40	21		21	66		13, 14
P26,Q26		2	8		3,5,6, 12,13, 19,20	24		24	91		13, 15
P27,Q27	Propagation delay time	5	8	Open	2,3,6, 12,13, 19,20	24	81	24	91	nS	13, 15

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient tTLH Min	Initial limits 25° ± 5°C ambient tTLH Max	Initial limits 25° ± 5°C ambient tTHL Min	Initial limits 25° ± 5°C ambient tTHL Max	Unit	Test fig
P28,Q28	Propagation delay time	12	8	Open	2,3,5,6, 13,19,20	24	81	24	91	nS	13, 15
P29,Q29		19	8		2,3,5,6, 12,13,20						
P30,Q30		21	27		22,24, 25,31, 32,39,40						
P31,Q31		24			21,22, 25,31, 32,39,40						
P32,Q32		31			21,22, 24,25, 32,39,40						
P33,Q33		39	27		21,22, 24,25, 31,32,40	24	81	24	91		
P34,Q34		16	37	Open	17	10	37	10	37		
P35,Q35		2	4	3	Open	3	23	3	22		
P36,Q36	Propagation delay time	5	7	6	Open	3	23	3	22	nS	13, 15

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient tTLH Min	Initial limits 25° ± 5°C ambient tTLH Max	Initial limits 25° ± 5°C ambient tTHL Min	Initial limits 25° ± 5°C ambient tTHL Max	Unit	Test fig
P37,Q37	Propagation delay time	12	14	13	Open	3	23	3	22	nS	13, 15
P38,Q38		19	18	20							
P39,Q39		21	23	22							
P40,Q40		24	26	25							
P41,Q41		31	33	32							
P42,Q42		34	36	35							
P43,Q43		39	38	40	Open	3	23	3	22		13, 15
P44,Q44		2	11	3,5	6	6	41	6	41		13, 14
P45,Q45		5	11	2,6	3						
P46,Q46		12	15	13, 19	20						
P47,Q47		19	15	12, 20	13						
P48,Q48	Propagation delay time	21	29	22, 24	25	6	41	6	41	nS	13, 14

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

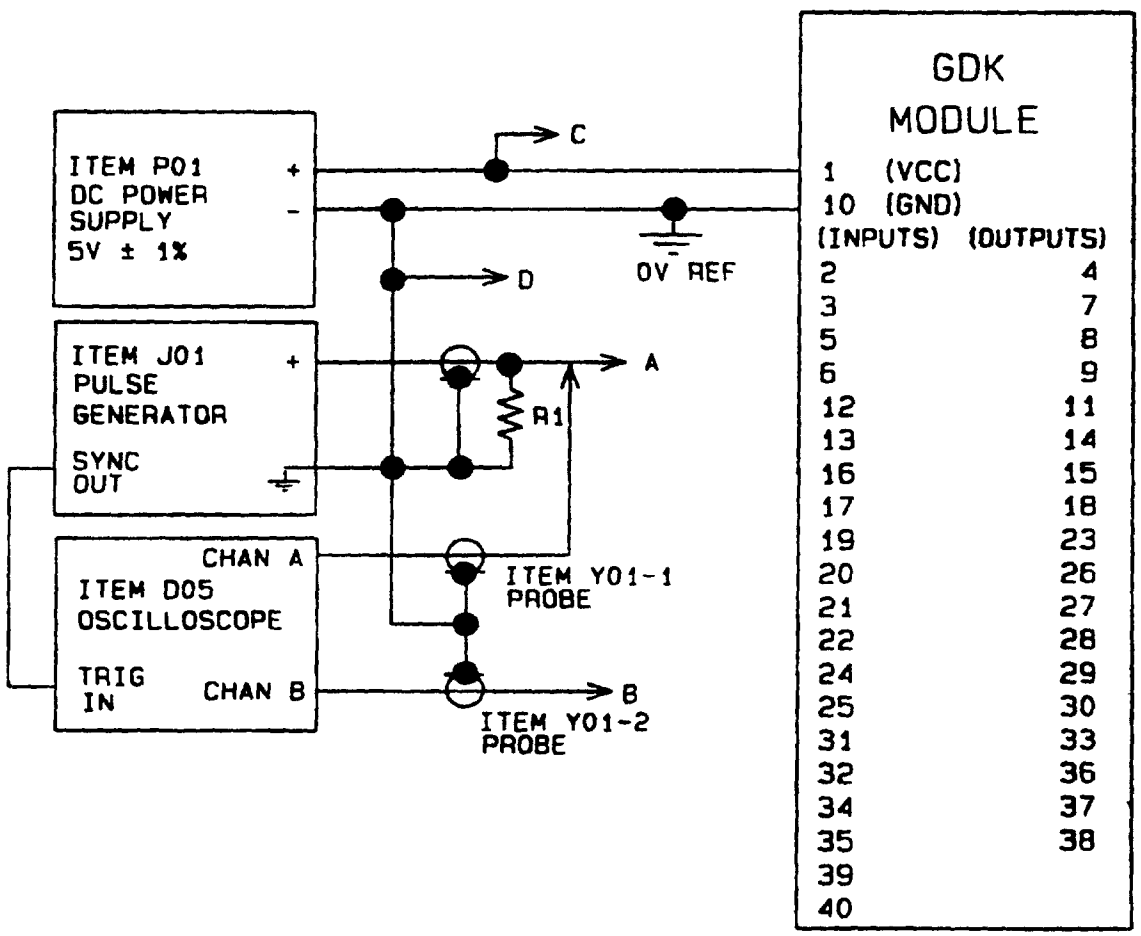
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient tTLH Min	Initial limits 25° ± 5°C ambient tTLH Max	Initial limits 25° ± 5°C ambient tTHL Min	Initial limits 25° ± 5°C ambient tTHL Max	Unit	Test fig
P49,Q49	Propagation delay time	24	29	21, 25	22	6	41	6	41	nS	13, 14
P50,Q50		31	30	32, 39	40						
P51,Q51		39	30	31, 40	32	6	41	6	41		13, 14
P52,Q52		2	9	3,5, 12	6,13,19, 20	9	64	9	59		13, 15
P53,Q53		5		2,6, 12	3,13,19, 20						
P54,Q54		12		2,13, 19	3,5,6,20						
P55,Q55		19	9	2,12, 20	3,5,6,13						
P56,Q56		21	28	22, 24,31	25,32, 39,40						
P57,Q57		24		21, 25,31	22,32, 39,40						
P58,Q58	Propagation delay time	31	28	21, 32,39	22,24, 25,40	9	64	9	59	nS	13, 15

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* TABLE XI. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test. - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient tTLH Min	Initial limits 25° ± 5°C ambient tTLH Max	Initial limits 25° ± 5°C ambient tTHL Min	Initial limits 25° ± 5°C ambient tTHL Max	Unit	Test fig
P59,Q59	Propagation delay time	39	28	21, 31,40	22,24, 25,32	9	64	9	59	nS	13, 15
P60,Q60		2	8	3,5, 12	6,13,19, 20	12	74	12	74		13, 14
P61,Q61		5		2,6, 12	3,13,19, 20						
P62,Q62		12		2,13, 19	3,5,6,20						
P63,Q63		19	8	2,12, 20	3,5,6,13						
P64,Q64		21	27	22, 24,31	25,32, 39,40						
P65,Q65		24		21, 25,31	22,32, 39,40						
P66,Q66		31		21, 32,39	22,24, 25,40						
P67,Q67		39	27	21, 31,40	22,24, 25,32	12	74	12	74		
P68,Q68	Propagation delay time	16	37	17	Open	6	33	6	33	nS	13, 14

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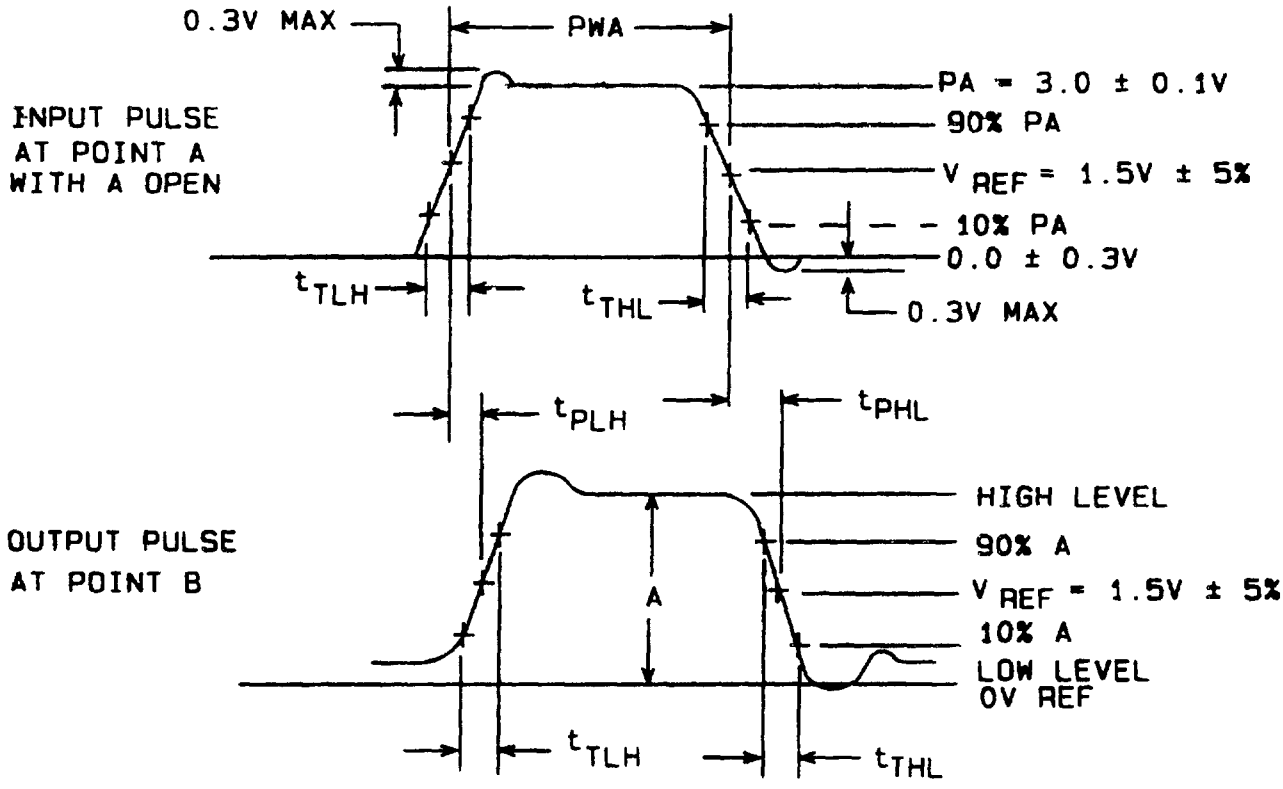
NOTES:

1. R1 = 50 Ohms ± 5 per cent, 1 W.
2. Connect load circuit number 1 (figure 9) to pins 8, 9, 27, 28, 36 and 37.
3. Connect load circuit number 2 (figure 9) to pins 4, 7, 11, 14, 15, 18, 23, 26, 29, 30, 33 and 38.
4. Before making the output measurements, the output levels shall be verified to be 0.4 maximum for low level and 2.4V minimum for high level.

*5. Input pins 2, 3, 5, 6, 12, 13, 16, 17, 19, 20, 21, 22, 24, 25, 31, 32, 34, 35, 39 and 40 shall be connected to point C, except when otherwise specified in table XI.

* FIGURE 13. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) time test circuit for table XI.

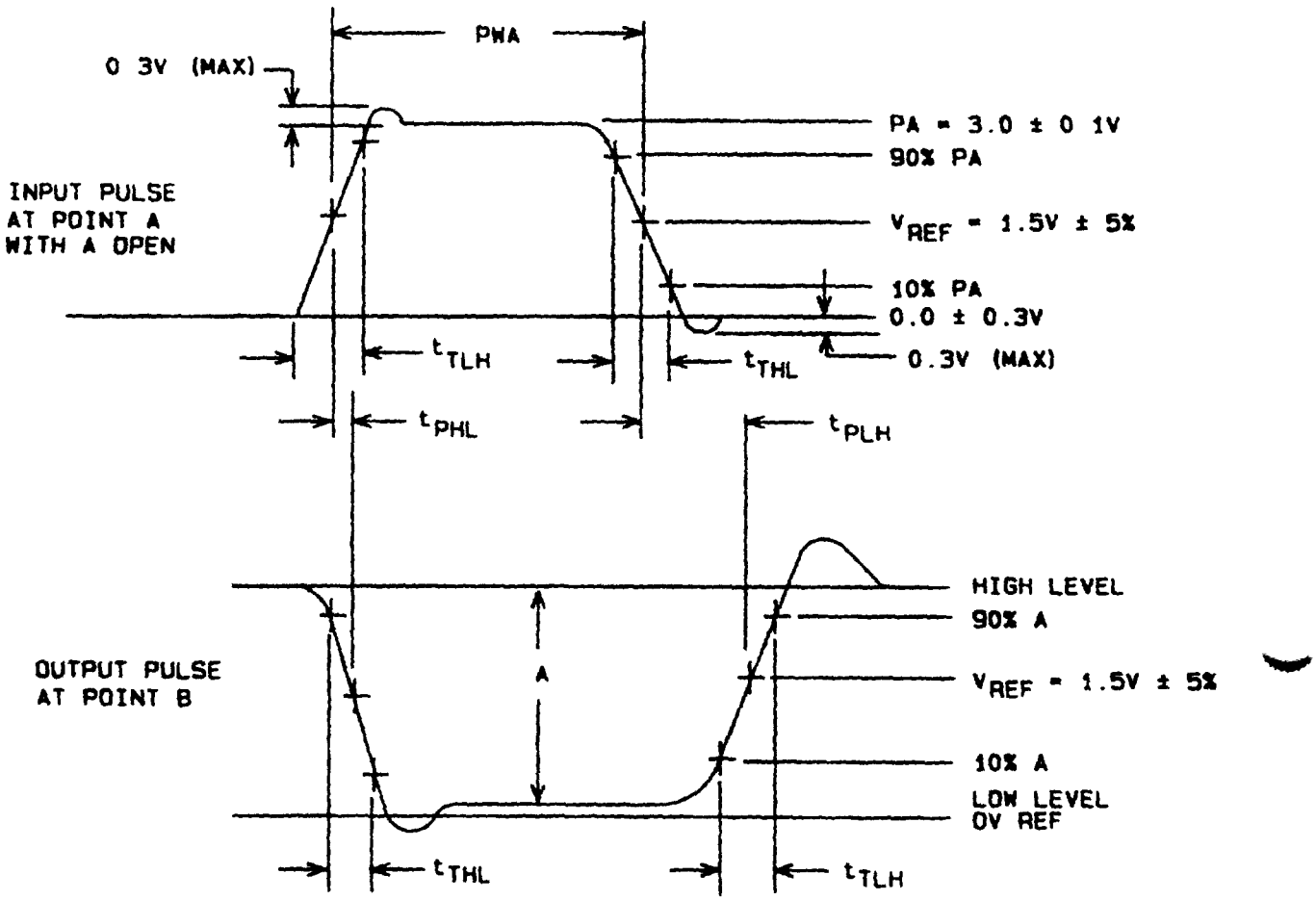
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NOTE: Input pulse parameters:
PWA = 200 nS ± 10 per cent.
tTLH = tTHL = 6 ± 1 nS.
PRR = 1 MHz ± 10 per cent.

FIGURE 14. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test circuit waveforms for table XI.

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NOTE: Input pulse parameters:
PWA = 200 nS ± 10 per cent.
tTLH = tTHL = 6 ± 1 nS.
PRR = 1 MHz ± 10 per cent.

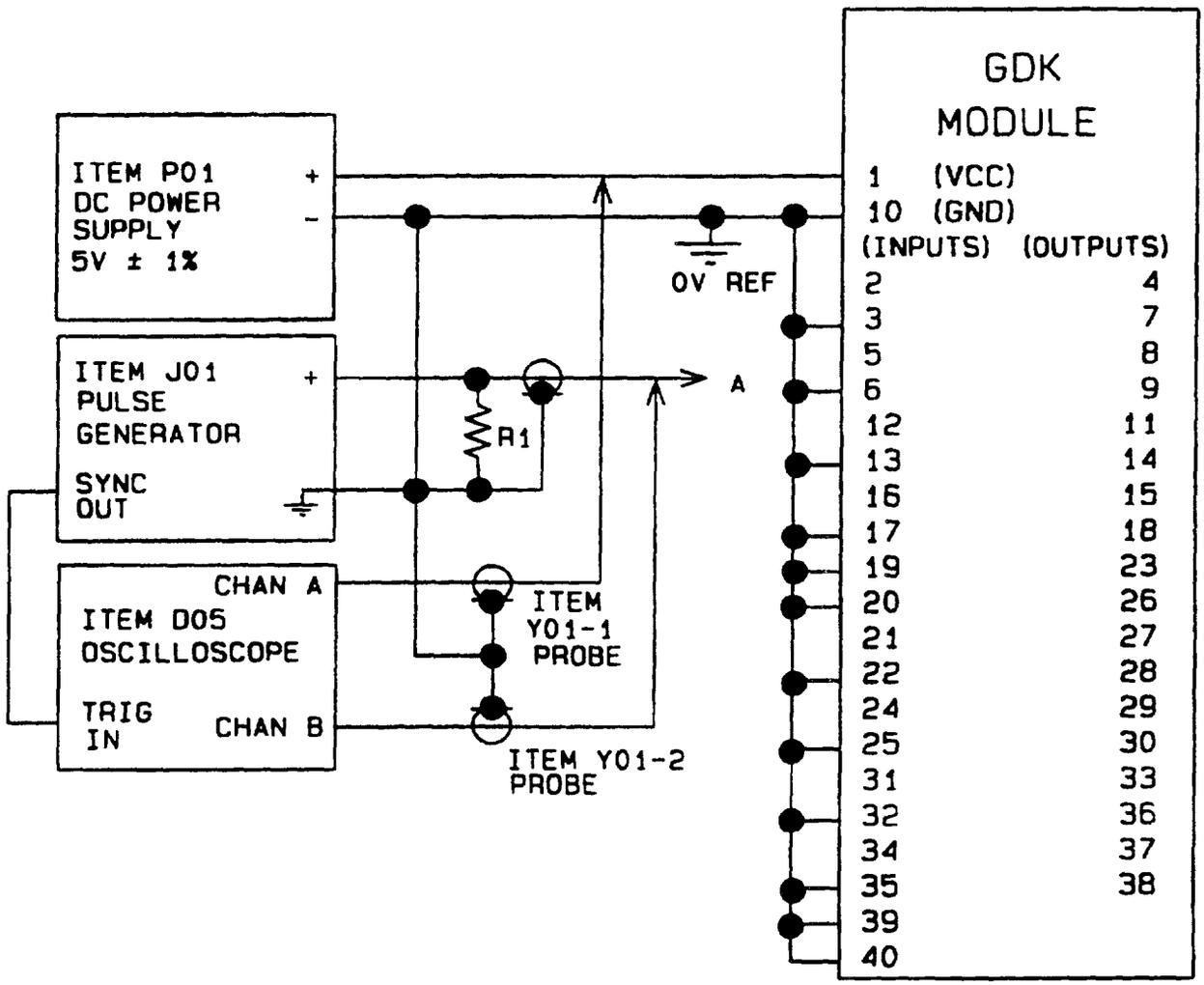
FIGURE 15. Transition time (tTLH and tTHL) and propagation delay time (tPHL and tPLH) test circuit waveforms for table XI.

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TABLE XII. Power supply transient amplitude test.

Test cond	Parameter	Connect	Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	Min	Max		
R	Power supply transient amplitude	2,5,12, 16,21, 24,31, 34	0	400	mV	16, 17

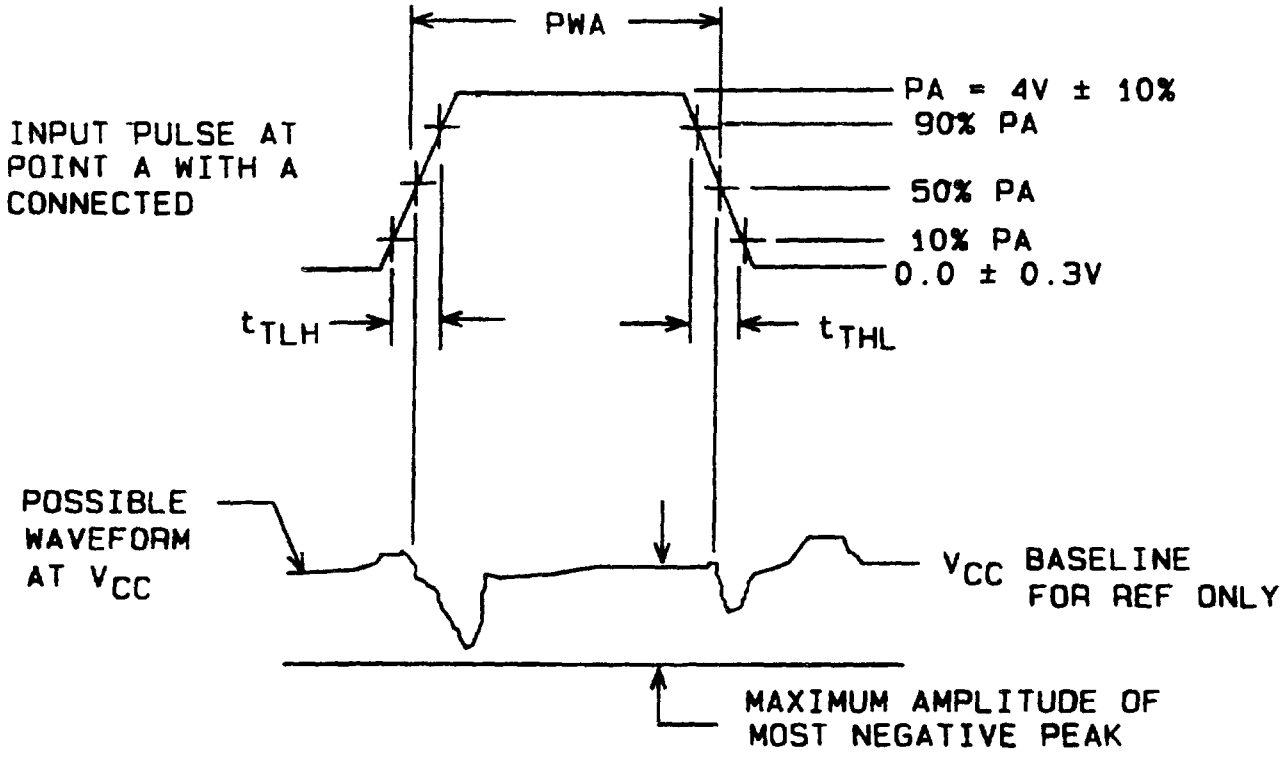
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NOTE: R1 = 50 Ohms ± 5 per cent, 1 W.

FIGURE 16. Power supply transient amplitude test circuit for table XII.

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NOTES:

1. **Input pulse parameters:**
   PWA = 3 μS ± 10 per cent.
   tTLH = tTHL = 10 ± 1 nS.
   PRR = 100 kHz ± 10 per cent.
2. **All voltages are referenced to ground.**

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XII.

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TABLE XIII. Output crosstalk (high level and low level) test

Test cond	Parameter	Connect							Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
S1	Output crosstalk, high level	2	3	12,16,19, 21,24,31, 34,39	4	Open	4	13,17,20, 22,25,32, 35,40	0	400	mV	18, 19
S2		5	6	12,16,19, 21,24,31, 34,39	7		7	13,17,20, 22,25,32, 35,40				
S3		2	3,5,6	16,21,24, 31,34,39	11		11	17,22,25, 32,35,40				
S4		12	13	2,5,16, 21,24,31, 34,39	14		14	3,6,17, 22,25,32, 35,40				
S5		19	20	2,5,16, 21,24,31, 34,39	18		18	3,6,17, 22,25,32, 35,40				
S6		12	13,19, 20	16,21,24, 31,34,39	15	Open	15	17,22,25, 32,35,40				
S7		2	3,5,6, 12,13, 19,20	16,21,24, 31,34,39	9	9	Open	17,22,25, 32,35,40				
S8		Open	2,3,5, 6,12, 13,19, 20	16,21,24, 31,34,39	8	8	Open	17,22,25, 32,35,40				
S9	Output crosstalk, high level	Open	16,17	2,5,12, 19,21,24, 31,34,39	37	37	Open	3,6,13, 20,22,25, 32,35,40	0	400	mV	18, 19

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TABLE XIII Output crosstalk (high level and low level) test - Continued.

Test cond	Parameter	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25° ± 5°C ambient Min	Max	Unit	Test fig
S10	Output crosstalk, high level	21	22	2,5,12, 16,19,31, 34,39	23	Open	23	3,6,13, 17,20,32, 35,40	0	400	mV	18, 19
S11		24	25	2,5,12, 16,19,31, 34,39	26		26	3,6,13, 17,20,32, 35,40				
S12		21	22,24, 25	2,5,12, 16,19,34	29		29	3,6,13, 17,20,32, 35				
S13		31	32	2,5,12, 16,19,21, 24,34	33		33	3,6,13, 17,20,22, 25,35				
S14		39	40	2,5,12, 16,19,21, 24,34	38		38	3,6,13, 17,20,22, 25,35				
S15		31	32,39, 40	2,5,12, 16,19,34	30	Open	30	3,6,13, 17,20,35				
S16		21	22,24, 25,31, 32,39, 40	2,5,12, 16,19,34	28	28	Open	3,6,13, 17,20,35				
S17		Open	21,22, 24,25, 31,32, 39,40	2,5,12, 16,19,34	27	27		3,6,13, 17,20,35				
S18	Output crosstalk, high level	34	35	2,5,12, 16,19,24, 31,39,21	36	36	Open	3,6,13, 17,20,22, 25,32,40	0	400	mV	18, 19

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TABLE XIII. Output crosstalk (high level and low level) test - Continued.

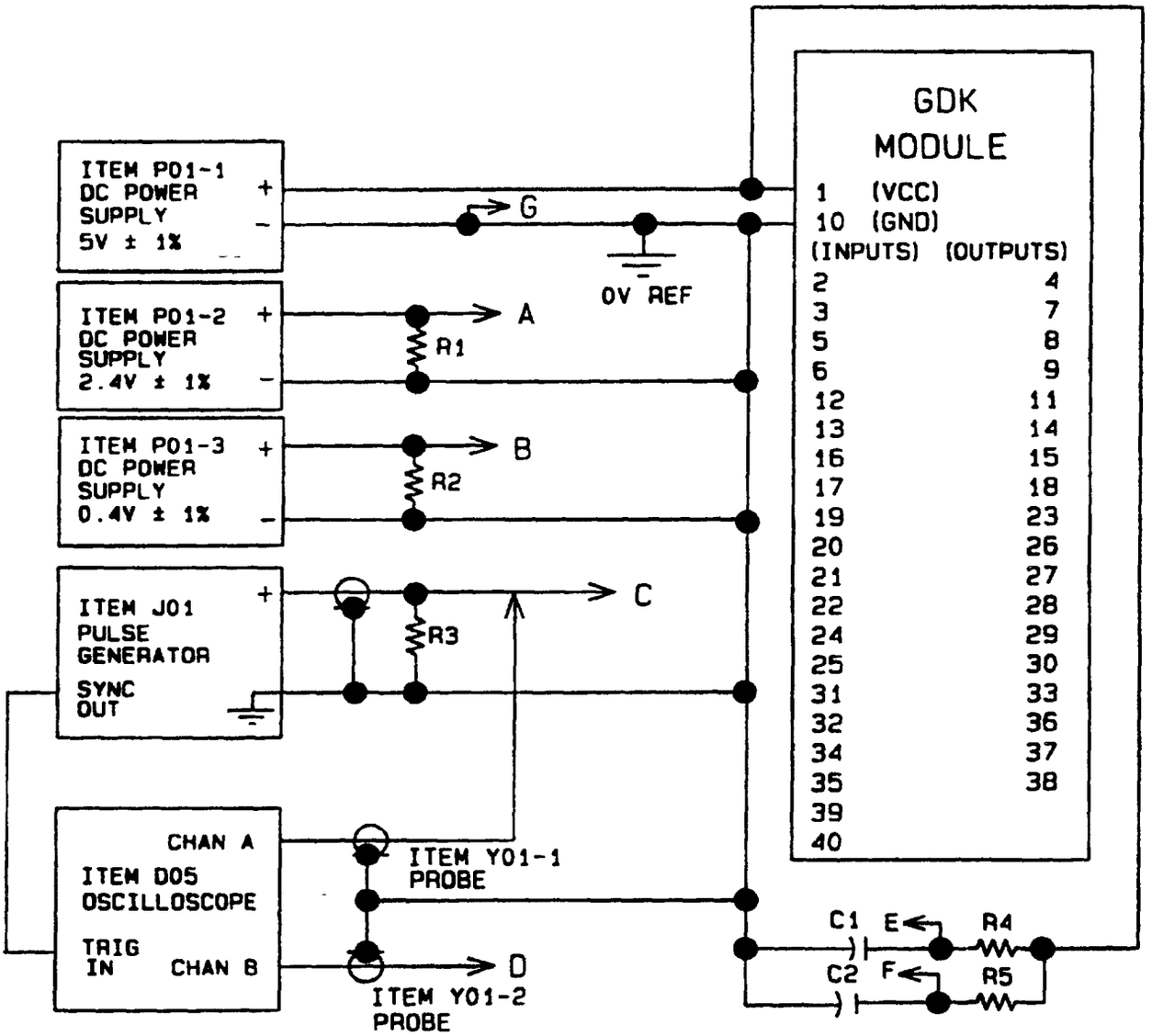
Test cond	Parameter	Connect							Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
T1	Output crosstalk, low level	Open	2,3	12,16,19, 21,24,31, 34,39	4	Open	4	13,17,20, 22,25,32, 35,40	0	400	mV	18, 19
T2			5,6	12,16,19, 21,24,31, 34,39	7		7	13,17,20, 22,25,32, 35,40				
T3			2,3,5, 6	16,21,24, 31,34,39	11		11	17,22,25, 32,35,40				
T4			12,13	2,5,16, 21,24,31, 34,39	14		14	3,6,17, 22,25,32, 35,40				
T5			19,20	2,5,16, 21,24,31, 34,39	18		18	3,6,17, 22,25,32, 35,40				
T6			12,13, 19,20	16,21,24, 31,34,39	15	Open	15	17,22,25, 32,35,40				
T7		Open	2,3,5, 6,12, 13,19, 20		9	9	Open					
T8		2	3,5,6, 12,13, 19,20	16,21,24, 31,34,39	8	8		17,22,25, 32,35,40				
T9	Output crosstalk, low level	16	17	2,5,12, 19,21,24, 31,34,39	37	37	Open	3,6,13, 20,22,25, 32,35,40	0	400	mV	18, 19

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TABLE XIII. Output crosstalk (high level and low level) test - Continued.

Test cond	Parameter	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
T10	Output crosstalk, low level	Open	21,22	2,5,12, 16,19,31, 34,39	23	Open	23	3,6,13, 17,20,32, 35,40	0	400	mV	18, 19
T11			24,25	2,5,12, 16,19,31, 34,39	26		26	3,6,13, 17,20,32, 35,40				
T12			21,22, 24,25	2,5,12, 16,19,31, 34	29		29	3,6,13, 17,20,32, 35				
T13			31,32	2,5,12, 16,19,21, 24,34	33		33	3,6,13, 17,20,22, 25,35				
T14			39,40	2,5,12, 16,19,21, 24,34	38		38	3,6,13, 17,20,22, 25,35				
T15			31,32, 39,40	2,5,12, 16,19,34	30	Open	30	3,6,13, 17,20,35				
T16		Open	21,22, 24,25, 31,32, 39,40		28	28	Open					
T17		21	22,24, 25,31, 32,39, 40	2,5,12, 16,19,34	27	27		3,6,13, 17,20,35				
T18	Output crosstalk, low level	Open	34,35	2,5,12, 16,19,21, 24,31,39	36	36	Open	3,6,13, 17,20,22, 25,32,40	0	400	mV	18, 19

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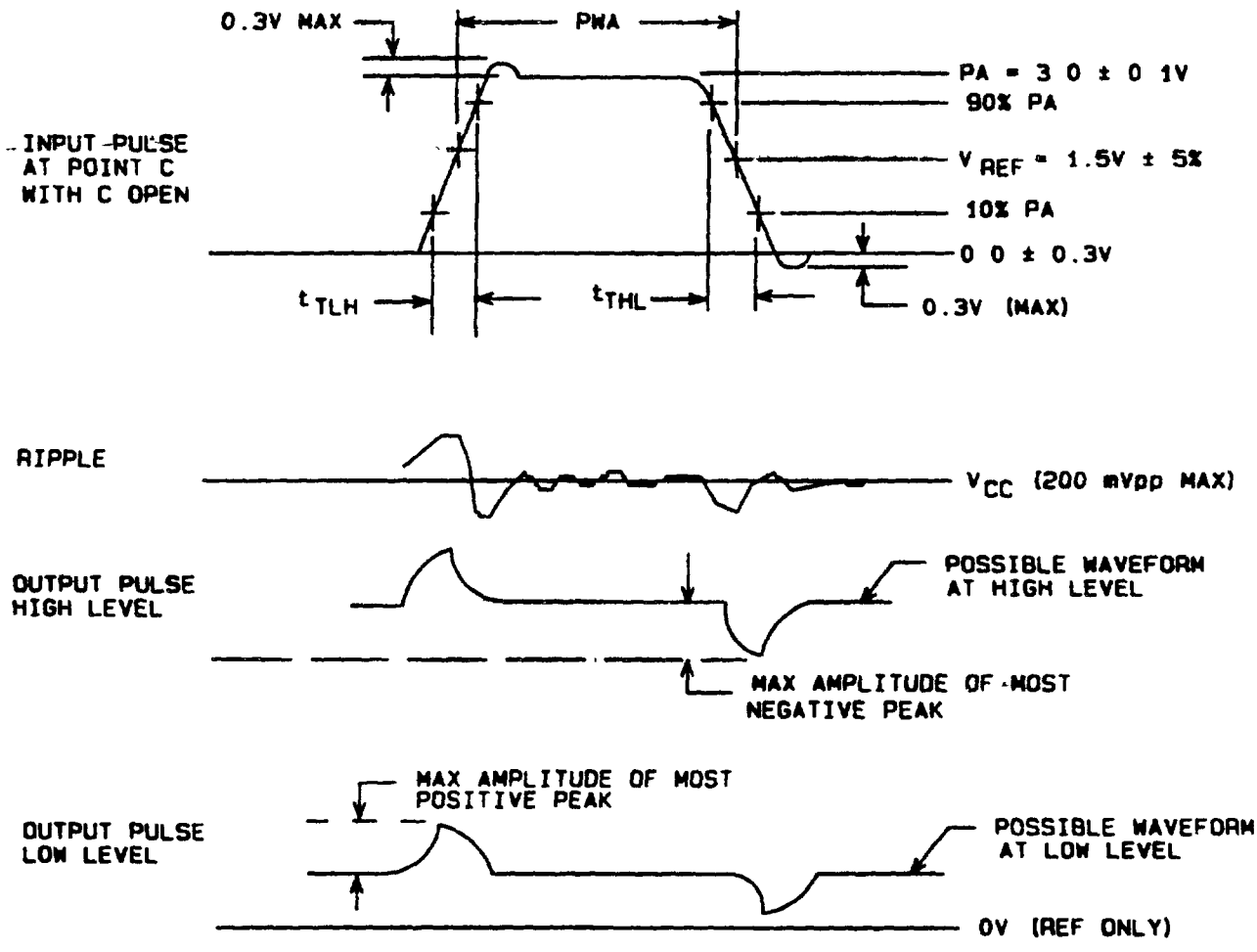


NOTES:

1. R1 = R2 = R3 = 50 Ohms ± 5 per cent, 1 W.
2. R4 = 316 Ohms ± 1 per cent, 1/8 W.
3. R5 = 374 Ohms ± 1 per cent, 1/8 W.
4. C1 = C2 = 30 ± pF (including probe capacitance).

FIGURE 18. Output crosstalk (high level and low level) test circuit for table XIII.

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NOTE: Input pulse parameters:
PWA = 200 nS ± 1 per cent.
tTLH = tTHL = 25 ± 5 nS.
PRR = 1 MHz ± 10 per cent.

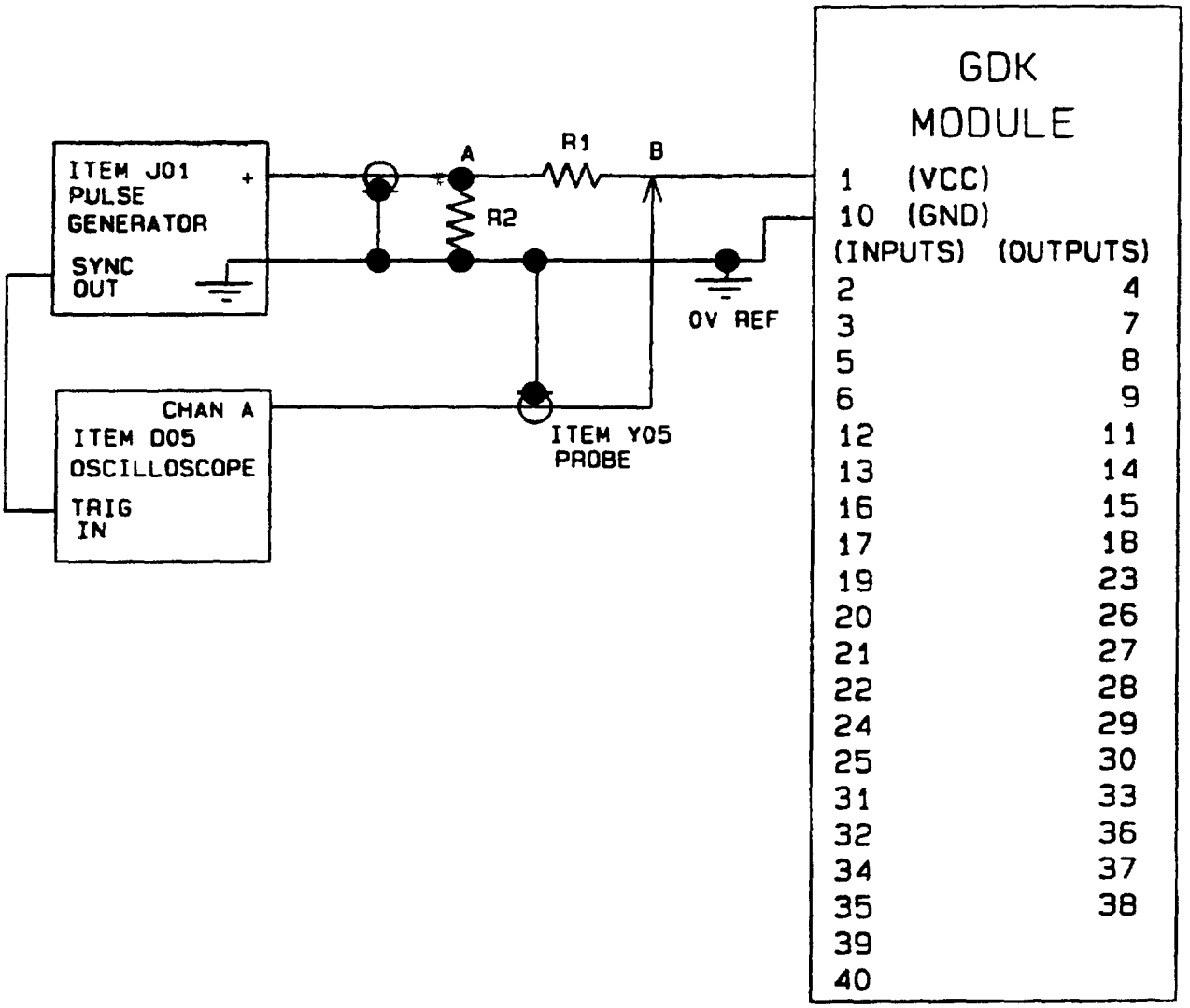
FIGURE 19. Output crosstalk (high level and low level) test circuit waveforms for table XIII.

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TABLE XIV. Power supply transient rise time test.

Test cond	Parameter	Initial limits 25 ± 5°C ambient		Unit	Test fig
		Min	Max		
U	Power supply transient rise	2.5	7.5	µS	20, 21

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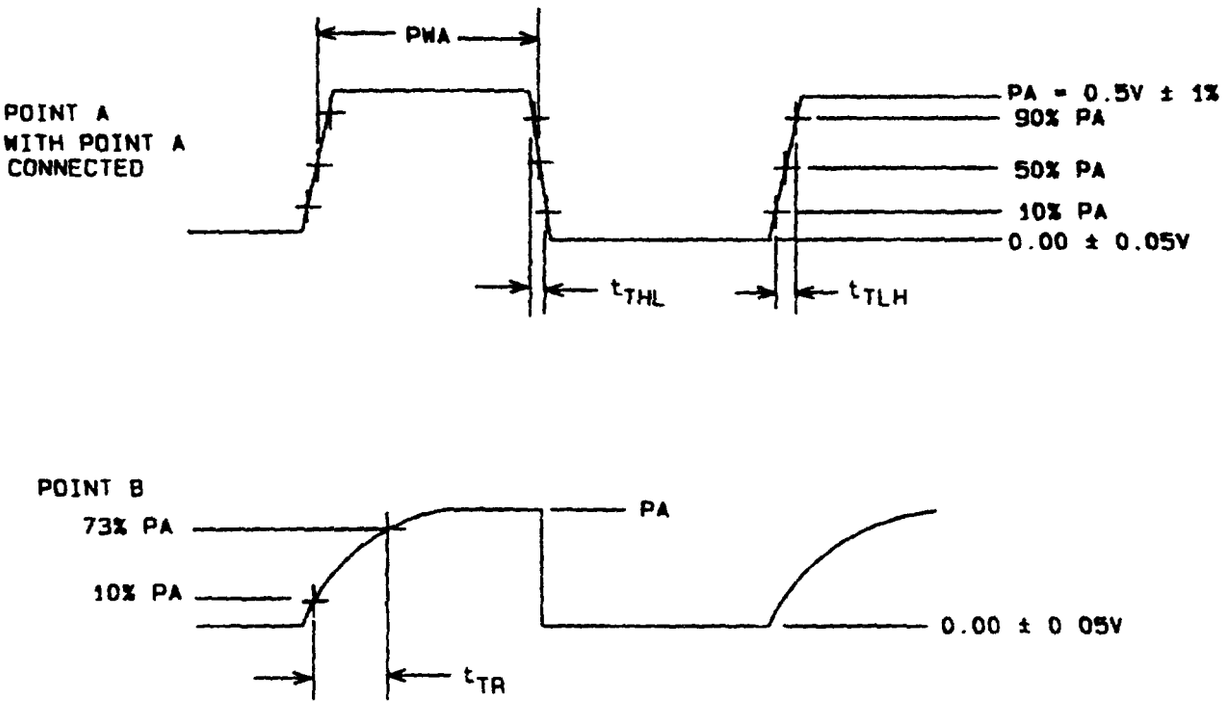


NOTES:

1. R1 = 412 Ohms ± 1 per cent, 1/8 W.
2. R2 = 50 Ohms ± 5 per cent, 1 W.

FIGURE 20. Power supply transient rise time test circuit for table XIV.

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NOTE: Input pulse parameters:
* PWA = 750 µS ± 10 per cent.
tTLH = tTHL = 50 nS ± 10 per cent.
* PRP = 1.5 mS ± 10 per cent.

* FIGURE 21. Power supply transient rise time test circuit waveforms for table XIV.

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